

SMART INDIA HACKATHON 2026 · FINAL TECHNICAL REPORT

StormLife Nowcast

Satellite-first convective-scale nowcasting of deep convection for India (0–6 h): implementation, validation and replay prototype

Problem Statement ID	SIH26084
Problem Statement title	Convective Scale Nowcasting for Thunderstorms, Hail & Cloudbursts (0–6 hr)
Organisation	Ministry of Earth Sciences (MoES) / NCMRWF
Category / Theme	Software / Disaster Management
Team name	[TEAM NAME]
Team ID	[TEAM ID]
Team leader	[TEAM LEADER]
Team members	[MEMBER 1] · [MEMBER 2] · [MEMBER 3] · [MEMBER 4] · [MEMBER 5]
Institution	[INSTITUTION / COLLEGE], [CITY, STATE]
Mentor(s)	[MENTOR NAME(S)]
Report status	Documents Phases 1–9, evidence frozen at model version ml-v0
Date	[SUBMISSION DATE]

Research prototype. Not an official IMD warning. All results are replays of observed satellite data on held-out days; no live or operational forecasting is claimed.

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How to read this report.

- Numbers are copied from frozen project files; the source is named beside each important value.
- Results are labelled **OBSERVED RESULT** (a measured number), **INTERPRETATION** (what we think it means) or **LIMITATION** (what it does not show).
- Items marked **FUTURE** or **NOT IMPLEMENTED** do not exist in the prototype.

2. Executive summary

Problem. SIH26084 asks for 0–6 h nowcasting of thunderstorms, hail and cloudbursts at 1–3 km, from radar, satellite and lightning data, shown on a GIS dashboard. In practice there is no open, live Indian radar or lightning feed that a student team can train on or verify against:

- IMD Doppler-radar raw data needs a licence;
- the IMD AWS/ARG portal was closed to the public in May 2025;
- general MOSDAC users receive INSAT data with a 3-day latency.

Approach: satellite-first. We built the whole system around the one source that was openly obtainable: the **NOAA/NCEP/CPC Globally Merged IR** product (single ~11 μm channel, 4 km, 30 min). It is used as an approved fallback for Meteosat-9/INSAT. We forecast one verifiable target: **where the cloud top will be colder than 235 K (deep convection)**, 30 min to 6 h ahead, on a 2 km grid over NW India (26–34° N, 72–84° E).

Storm detection and tracking.

- Cold-cloud objects are detected with tobas.
- The first tracks were audited before any cell-level result was used, and failed. Direction reversals were 47 %, and 20.6 % of cold links were likely identity switches.
- They were replaced by overlap-based tracking with a coldness-weighted centroid, which cut likely identity switches to 0.4 %.

Baseline forecasting. Persistence and pySTEPS motion extrapolation were scored on three held-out hail-report days: 14 May (E8), 4 May (E10) and 16 May 2026 (E11).

- pySTEPS beats persistence at **every lead on all three days**: pooled CSI 0.586 vs 0.511 at 30 min, and 0.258 vs 0.176 at 2 h.
- Useful skill lasts about **90 min (persistence) and 120 min (pySTEPS)**.
- An exact error decomposition shows that pySTEPS' apparent over-forecasting at long leads is mainly a scoring-edge effect, not ignored storm decay.
- A decay-aware extension was stopped by a pre-declared rule, because the cell lifecycle trends did not persist.

ML probability prototype (ml-v0).

- **Model:** a histogram gradient-boosting classifier with isotonic calibration.
- **Data:** trained on 14 days (18–31 May 2026) and calibrated on 7 days (6–12 May 2026); the three event days were held out.
- **Probability skill:** the best of all methods out to about 4 h. Brier skill score is 0.67 / 0.51 / 0.27 / 0.13 at 30 min / 1 h / 2 h / 3 h, against 0.63 / 0.41 / 0.02 / –0.23 for a no-fit neighbourhood pySTEPS probability.
- **Calibration:** good, with calibration error 0.002–0.004.
- **As a yes/no forecast at $p \geq 0.5$** it does **not** beat pySTEPS beyond 1 h.
- **Mechanism:** it behaves mainly as a calibrated, lead-aware smoothing of pySTEPS. Storm-initiation skill is not demonstrated.

Replay, validation and decision support.

- **Evidence freeze:** all evidence (386 files) is frozen with SHA-256 hashes, and a test fails if any of it changes.
- **Web prototype:** a static React + TypeScript + MapLibre app replays the frozen outputs across nine screens: event selection, situation, storm cells, 0–6 h forecast, alert center (exercise), historical replay, confidence, model performance and data health.
- **Replay view:** three synced maps show what the system knew, what it predicted and what actually happened.
- **Labelling:** every screen is marked REPLAY; the ML output is shown as a probability with its validated skill.

Limits.

- Only 3 held-out days, all in May 2026 over NW India, and the ML test is backward in time.

- The input is a 30-min, single-channel IR source, with unverified parallax.
- There is no independent radar or lightning truth, and no live feed.

3. Problem statement

The official statement (SIH portal, PS ID **SIH26084**, MoES/NCMRWF; transcribed in the research report, Part 1) asks teams to:

- "Build a real-time, convective-scale Nowcasting System (0–6 hour lead time) operating at a hyper-local 1–3 km spatial resolution";
- design a system "rooted in Multi-Source Data Fusion architectures";
- "ingest high-frequency, heterogeneous meteorological streams, automatically detect early convective initiation, and dynamically forecast severe storm parameters".

The expected inputs are Doppler weather radar (reflectivity and velocity), geostationary satellite imagery (INSAT-3D/3DR thermal/IR bands) and ground-based lightning networks. The expected output is "a real-time, interactive GIS-mapped dashboard showcasing high-resolution (1–3 km) hazard zones with live countdown clocks for storm arrivals."

Requirement (PS)	What this project delivers	Status
0–6 h lead time	Forecasts at 30, 60 ... 360 min (30-min steps)	Implemented; useful skill $\approx$ 2 h (baselines), BSS > 0 to $\approx$ 4 h (ML)
1–3 km resolution	2 km common grid (effective $\approx$ 4 km, source resolution)	Implemented
Multi-source fusion (radar, satellite, lightning)	Satellite IR only (NOAA CPC merged IR)	Partial: radar and lightning not openly available
Convective initiation detection	Not demonstrated	Not implemented
Thunderstorm / deep-convection nowcast	BT < 235 K probability and deterministic forecasts	Implemented and validated on 3 days
Lightning density, hail probability, downburst velocity, cloudburst thresholds	No hazard-specific truth available	Not implemented; no skill claimed
GIS dashboard	React + MapLibre replay prototype, 9 screens	Implemented (replay only)
Live countdown clocks / real time	Replay of observed data; latency not modelled	Not implemented (future)
Users: local administrations, aviation, farming	Exercise advisories (CAP 1.2, status=Exercise)	Prototype only; alert skill not verified

Target hazards and the scope decision. The PS lists thunderstorms, lightning, hail, downbursts and cloudbursts. We scoped the verifiable part to **deep convection**, defined as cloud-top brightness temperature (BT) below 235 K (VALIDATION_PLAN §2). It can be verified from the satellite field itself. The other hazards need truth data (radar, lightning, hail pads, rain gauges) that were not openly available. They are treated as future work, and no skill is claimed for them.

4. Motivation and background

Why 0–6 h convective forecasting is hard. Convective cells form, grow and decay within tens of minutes to a few hours. Extrapolation methods (moving the current field along its motion) cannot create new storms or model decay, so their skill falls quickly with lead time. The PRD expected skill to decay after about 1–2 h, and the project measured this directly: useful extrapolation skill ends at about 2 h (Section 17).

Why the data situation matters in India (research report Part 6, verified at the time of writing):

- **Radar:** IMD Doppler weather radar (DWR) raw volumes need a licence or MoES research request (DATA_REALITY: *UNAVAILABLE / RESTRICTED*).
- **Station data:** the IMD AWS/ARG portal was locked to the public in May 2025.

- **INSAT via MOSDAC:** registered general users get limited datasets with a **3-day latency**; near-real-time access is for privileged users only.
- **Lightning:** Blitzortung's terms forbid use in storm-warning systems. ISS-LIS ended on 16 Nov 2023. NRSC and IITM networks need research access.
- **Operational nowcasting:** IMD's radar-based nowcasts (WDSS-II) cover radar-served cities, 156 per the research report.

A design that depends on radar or lightning cannot be built and verified openly. A **satellite-first** design can.

Existing operational context. Storm-lifecycle and cell-hazard systems already exist: NWCSAF RDT-CW, MeteoSwiss TRT and NOAA ProbSevere v3 (research report Part 9; REVIEW_AND_SCOPE R6). The concept is therefore **not novel**. The project's contribution is a verified, reproducible, satellite-only implementation on Indian storm days, with its limits stated (Section 7).

5. Objectives

#	Objective (measurable)	Outcome
O1	Build a reproducible real-data pipeline: download → QC → 2 km grid → cells, with SHA-256 provenance	Achieved. Byte-identical reruns; raw-file manifest with URL, Last-Modified, size, SHA-256
O2	Detect and track convective cells, and audit tracking quality before using cell-level metrics	Achieved. v1 tracks rejected by the audit; v2 overlap tracking cut ID-switch candidates 0.206 → 0.004 (E8)
O3	Produce and verify persistence and pySTEPS forecasts at 30–360 min	Achieved. 45 issue times per day, 12 leads, CSI/POD/FAR/bias/FSS
O4	Test whether baseline findings generalise to 2–3 more real events, with uncertainty	Achieved. E10 and E11 added (E8a excluded by rule); block-bootstrap intervals
O5	Explain the remaining forecast error and test a decay-aware extension	Diagnosis achieved. Extension stopped by a pre-declared reliability rule
O6	Train a lightweight, leakage-safe ML model and evaluate it once on held-out days against the baselines	Achieved. ml-v0 evaluated on E8/E10/E11; calibrated probabilities
O7	Freeze all evidence and guard it with tests	Achieved. 386 files hashed; freeze-guard test
O8	Deliver a working replay prototype of the dashboard	Achieved. 9 screens; build, unit and browser tests pass

6. Proposed solution

StormLife Nowcast is a pipeline that turns raw geostationary IR frames into verified forecasts and a replayable decision-support view:

1. **Ingest** NOAA CPC merged-IR hourly files (two half-hour fields each). Every file is recorded in a manifest with its SHA-256.
2. **Quality-control and regrid.** Decode brightness temperature, fill tiny gaps (≤ 4 native pixels, flagged), and regrid onto a 2 km Lambert azimuthal equal-area (LAEA) grid of 453×603 pixels. Each frame gets a QC status.
3. **Detect and track cells.** tobac detects cold-cloud features and segments cells colder than 245 K. The v2 tracker links cells frame to frame by area overlap and assigns a coldness-weighted centroid. Lifecycle diagnostics include min BT, area, cold-core area, ΔBT and phase rules v0 (proposed).
4. **Forecast** the 2 km BT field 30–360 min ahead with three methods:
 - **persistence** (no change);
 - **pySTEPS** (Lucas–Kanade motion + semi-Lagrangian advection);
 - **ML (ml-v0):** a calibrated probability that $BT < 235$ K.
5. **Verify** every method on identical pixels of held-out days: CSI, POD, FAR, bias, FSS, Brier skill score and reliability, with block-bootstrap uncertainty.
6. **Freeze** the evidence (386 files, SHA-256).
7. **Build static replay bundles** from the frozen files only, checking each file's hash.

8. **Replay** the bundles in a React + MapLibre dashboard with honest labels. Forecasters can approve or reject exercise advisories (CAP 1.2, status=Exercise).

7. Key innovation / differentiation

These are the project's defensible differentiators. None is claimed to be universally novel. Operational systems with related concepts exist (Section 4).

Differentiator	What was actually done	Evidence
Satellite-first workflow	End-to-end system on openly obtainable geostationary IR, with no simulated radar or lightning	FIRST_EVENT; DATA_REALITY; Section 11
Reproducible real-event validation	Real hail-report days, selected by a rule fixed before scoring; 386 evidence files hash-frozen	MULTI_EVENT §1; FREEZE_ml-v0.json
Baseline-relative evaluation	Every method scored against persistence and pySTEPS on the same pixels. Negative results reported (yes/no ML, decay model stop)	BASELINE_EVAL; ML_PROTOTYPE; DECAY_AWARE
Lifecycle / tracking audit	Tracks audited before cell-level use; v1 rejected, v2 overlap tracking adopted	TRACKING_AUDIT
Calibrated ML probability prototype	Isotonic calibration on validation days only; reliability and ECE reported on held-out days	ML_PROTOTYPE §3
Uncertainty / data-health visibility	Per-frame QC, source status, unscorable fraction, within-event intervals shown in the UI	Data Health and Confidence screens
Historical replay	<i>Knew / predicted / happened</i> panes with a method toggle, using only data available at issue time	REPLAY_SPEC; PRODUCT_DEMO
Explainable decision-support interface	Every forecast shows its validated skill at that lead, evidence level and envelope. Advisories require human approval	UI_UX_SPEC; Section 19

8. Complete system workflow

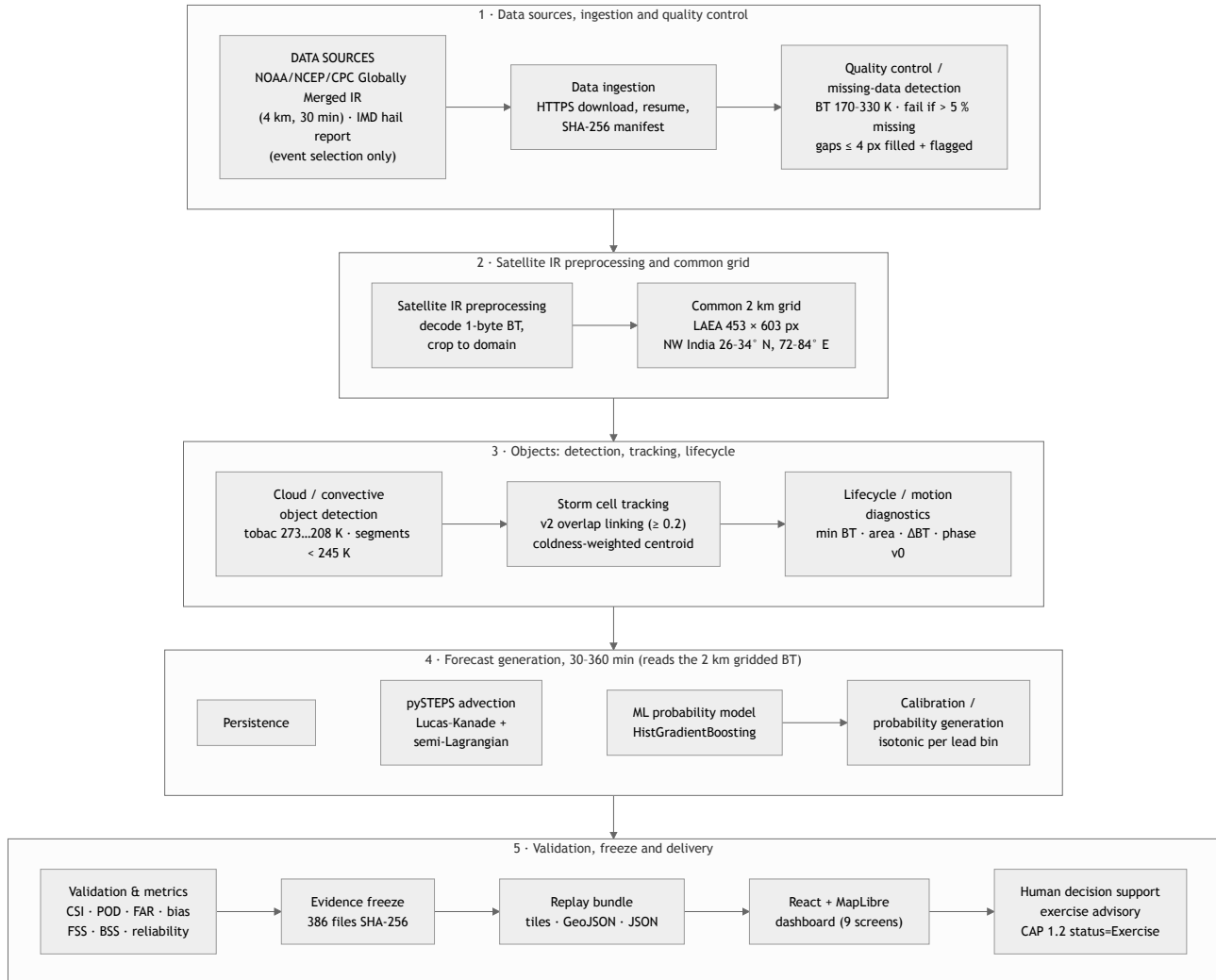


Figure 1 (Diagram 1). End-to-end system flowchart. Source: docs/DIAGRAMS.md .

9. System architecture

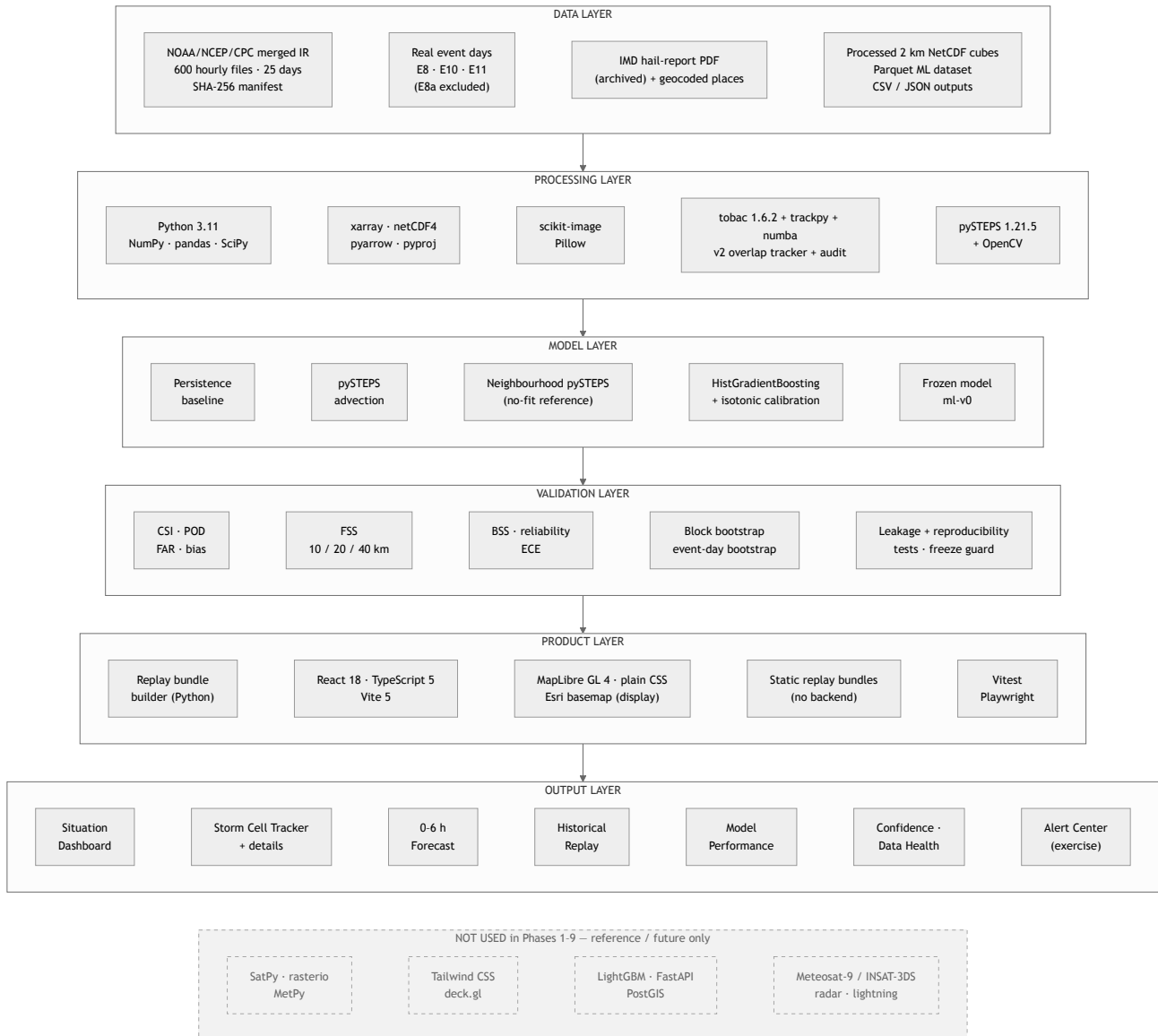


Figure 2 (Diagram 2). Layered architecture. Solid layers contain only technologies used in Phases 1–9. The dashed box lists technologies named in the PRD/ARCHITECTURE that were **not** used: SatPy, rasterio, MetPy, Tailwind CSS, deck.gl, LightGBM, FastAPI, PostGIS.

Layer notes:

- **Data layer.**
 - 600 hourly raw files over 25 days (the 4 event days E8/E8a/E10/E11 plus 21 training/validation days), about 16 GB.
 - Processed outputs are NetCDF cubes, Parquet ML rows and CSV/JSON metric tables.
- **Processing layer.**
 - Python 3.11.
 - tobac 1.6.2 uses trackpy for linking, with numba for speed.
 - pySTEPS 1.21.5 was built from the PyPI source with `–fopenmp` removed; there is no macOS arm64 wheel and Apple clang has no OpenMP. This affects only multithreading of code paths the project does not use.
- **Model layer.** Two deterministic baselines, a no-fit neighbourhood reference and one ML model (ml-v0), frozen.

- **Validation layer.** Metric code in `ml/verification/scores.py`. The FSS equals the pySTEPS implementation (tested).
- **Product layer.** A static site. There is no backend server and no database; the bundles are plain files.
- **Output layer.** Nine screens (Section 19).

10. Technology stack

Status legend:

- **Implemented:** code written in this project.
- **Used:** third-party library or service used at run time.
- **Reference:** named in project docs but not used.
- **Future:** planned, not built.

Category	Technology (version)	Purpose	Status
Programming	Python 3.11.16 (uv virtual environment)	All data, tracking, forecasting, ML and verification code	Used
Programming	TypeScript 5.9.3	Web prototype source	Used
Programming	Node.js 26.7.0 / npm 11.19.0	Web build tooling	Used
Data processing	NumPy 2.4.6 · pandas 3.0.6 · SciPy 1.17.1	Arrays, tables, filters (neighbourhood fractions, FSS)	Used
Data processing	xarray 2026.7.0 · netCDF4 1.7.4	Gridded cubes and forecast files (NetCDF)	Used
Data processing	pyarrow 25.0.1	Parquet (ML dataset, tobac features)	Used
Data processing	PyYAML 6.0.3 · requests 2.34.2	Configs; HTTPS download with resume	Used
Weather / scientific	pyproj 3.7.2	LAEA grid, reprojection to the display grid, geodesic motion	Used
Weather / scientific	pySTEPS 1.21.5 (local build) + OpenCV 5.0.0.93	Lucas–Kanade motion, semi-Lagrangian extrapolation, FSS cross-check	Used
Tracking	tobac 1.6.2 · trackpy 0.7 · numba 0.67.0	Multi-threshold feature detection, 245 K segmentation, v1 linking	Used
Tracking	v2 overlap tracker (<code>ml/tracking/overlap_link.py</code>)	Overlap linking, merge/split events, coldness centroid	Implemented
Tracking	scikit-image 0.26.0	Segmentation support (tobac) and contour extraction for map polygons	Used
ML	scikit-learn 1.9.1: <code>HistGradientBoostingClassifier</code> , <code>IsotonicRegression</code>	ML probability model and calibration	Used
ML	joblib 1.6.0	Model serialisation (frozen artifacts)	Used
Visualization	matplotlib 3.11.2	Scientific figures (figs 1–21)	Used
Visualization	Pillow 12.3.0	8-bit data tiles for the web bundles	Used
Frontend	React 18.3.1 · react-dom 18.3.1	Dashboard UI (9 screens)	Used
Frontend	Plain CSS with UI_UX_SPEC design tokens · IBM Plex fonts (Google Fonts)	Styling	Used
Mapping	MapLibre GL JS 4.7.1	Map, raster image source, GeoJSON cells/contours, synced replay panes	Used
Mapping	Esri World Light Gray Canvas tiles	Basemap (display only; can be switched off)	Used
Geocoding	OpenStreetMap Nominatim	Geocoding the IMD hail-report place names	Used
Storage	Files: NetCDF, Parquet, CSV, JSON, GeoJSON, PNG, joblib	All data and outputs; SHA-256 manifests	Used
Storage	PostGIS / any database	Planned in ARCHITECTURE	Future
Testing	pytest 9.1.1	77 Python tests (76 pass, 1 skip)	Used
Testing	Vitest 2.1.9	6 web unit tests	Used

Category	Technology (version)	Purpose	Status
Testing	Playwright 1.63.0 (headless Chromium)	2 end-to-end browser tests of the demo flow	Used
Build / deployment	Vite 5.4.21 · @vitejs/plugin-react 4.7.0	Type-checked production build and local preview	Used
Build / deployment	Docker, FastAPI, cloud hosting	Planned in ARCHITECTURE	Future
Reference only	SatPy, rasterio, MetPy, xskillscore	Named in ARCHITECTURE; not installed	Reference
Reference only	Tailwind CSS, deck.gl	Named in ARCHITECTURE/UI spec; plain CSS and MapLibre layers used instead	Reference
Reference only	LightGBM, PyTorch, SHAP	LightGBM replaced by scikit-learn's histogram GBDT (same family); no deep learning	Reference

Table 1. Technology stack. Versions come from the project environment (`requirements.txt`, `apps/web/package.json`, `docs/FREEZE_ml-v0.json`).

11. Data sources

Source	Data type	Time step	Grid	Period used	Usage	Limitations	Reproducibility
NOAA/NCEP/CPC Globally Merged IR (ftp.cpc.ncep.noaa.gov/precip/global_full_res_IR/)	Merged geostationary window-channel IR BT (~11 µm), 1-byte (BT – 75 K), 255 = missing	30 min (2 fields per hourly file)	4 km (regridded to 2 km)	1, 4, 6–12, 14, 16, 18–31 May 2026 (25 days, 600 hourly files ≈ 16 GB)	All model inputs and verification truth	Single channel; 30-min cadence; archive on the server starts 30–31 Mar 2026 then 1 May (no April 2026); satellite ID code 5 unverified; parallax status unverified	data/raw/cpc_merged_ir/manifest.json: URL, HTTP Last-Modified, bytes, SHA-256 per file; pipelines.inges t.cpc_merged_ir
CPC satellite-ID file (geomerg_satid_202605.Z)	Satellite contributing each pixel	Monthly file	4 km	May 2026	Diagnostic (one satellite, code 5, covers the domain)	Code-to-satellite mapping NEEDS VERIFICATION	Same manifest
IMD RMC New Delhi hail-storm report (PDF, updated 16 May 2026)	Hail occurrence by date and place	Daily (date only, no time)	Town names	Feb–May 2026	Event-day selection and map context only , never verification truth	Place/state column layout lost in text extraction; reports incomplete; no times	Archived as hails_torm_report_fac e59166ab1.pdf with SHA-256
OpenStreetMap Nominatim	Place coordinates	—	Point	—	Geocoding report places	Some names not found (Shirsi; Jubbarhatti, Hindon AFS; Tungnath)	Cached in data/raw/hail_reports/geocoded_*.json
Esri World Light Gray Canvas	Basemap tiles	—	Web tiles	—	Display context only	Terms for public deployment and boundary depiction NEEDS VERIFICATION	Online; can be switched off
Meteosat-9 IODC SEVIRI	Multi-channel 15-min imagery	15 min	~3 km nadir	—	Planned primary	Not connected (no EUMETSAT token)	—
INSAT-3DR / 3DS (MOSDAC)	Multi-channel imagery	30 min (15 min staggered)	1–8 km by channel	—	Planned	Not connected; 3-day latency for general users	—
IMD DWR radar; lightning networks	Reflectivity; flashes	—	—	—	Planned independent truth	Unavailable / restricted	—

Table 2. Data sources. The only data inputs are the CPC merged IR frames.

Source substitution.

- The approved satellite routes in DATA_REALITY (Meteosat-9 IODC via EUMETSAT; INSAT via MOSDAC) need credentials that were not available. EUMETSAT download returned HTTP 404 without a token, and there was no MOSDAC account.
- The CPC merged-IR product was therefore adopted in Phase 1 as an approved fallback. It was kept for all later phases, where the instructions were "keep NOAA IR as the approved Phase-1 fallback; do not acquire new satellite sources".
- Consequences: no water-vapour, visible or tri-spectral channels, and a 30-min cadence instead of 15 min.

- Long-term archive access for older years (e.g. NASA GES DISC GPM_MERGIR, which needs an Earthdata login) was not verified.

12. Data pipeline

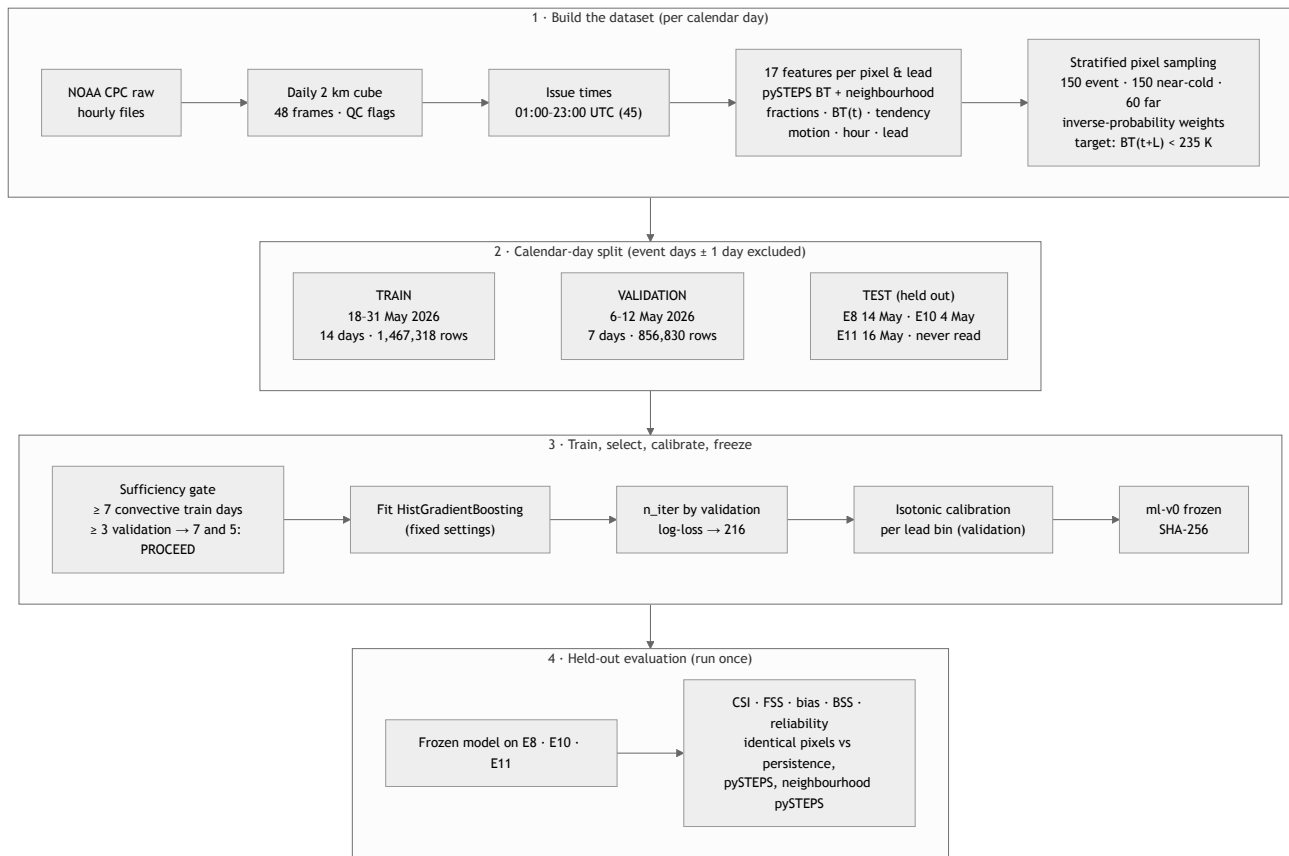


Figure 3 (Diagram 3). Data / ML pipeline, including the calendar-day splits used for the ML prototype.

Step	Implementation	Settings (frozen)
Raw satellite data	Hourly .Z files, two half-hour fields each; parallel HTTPS download with resume and file-locked manifest	Resume via HTTP Range; total size checked
Decoding	1-byte value + 75 $\rightarrow$ BT in K; 255 $\rightarrow$ missing; rows north $\rightarrow$ south	<code>pipelines/preprocess/decode.py</code>
Missing-data handling	Source gaps of ≤ 4 connected native pixels filled from valid 8-neighbours and flagged in a <code>gapfilled</code> mask; larger gaps stay missing	E8: 0.8–6.1 % missing per frame before fill, 0.6–5.4 % after
Projection / grid	NaN-aware bilinear regridding (normalised weights, minimum weight 0.5) onto LAEA (centre 30° N, 78° E), 2 km	453 × 603 px; <code>in_domain</code> mask for 26–34° N, 72–84° E; effective resolution $\approx$ 4 km
Frame QC	Status OK if in-domain missing $\leq 5\%$ and all BT in 170–330 K; otherwise QC_FAIL	Failed frames are kept and flagged , not dropped (E8: 6, E10: 9, E11: 0)
Cloud-top temperature	BT field per 30-min frame (48 frames per day, 00:00–23:30 UTC)	—
Convective thresholds	Deep convection = BT < 235 K (forecast target); cell segmentation < 245 K; cold core < 221 K	VALIDATION_PLAN §2; <code>config/tracking.yaml</code> ; <code>config/phases.yaml</code>
Object detection	tobac multi-threshold features at 273 / 253 / 235 / 221 / 208 K	Minimum size 25 px at 273 K, 8 px at 253 K, 4 px colder
Tracking	v1 tobac/trackpy (rejected by audit); v2 overlap linking of < 245 K segments (adopted)	Section 13

Step	Implementation	Settings (frozen)
Forecast input	2 km BT cube (grid methods); cell tables (object diagnostics, dashboard)	Issue times 01:00–23:00 UTC: 45 per day

Table 3. Data-pipeline steps and settings. Sources: FIRST_EVENT, BASELINE_EVAL, `config/*.yaml`.

LIMITATION: parallax. Missing pixels line cloud edges. That pattern is consistent with holes left by a parallax shift in the merged product. Whether the product is already parallax-corrected is **not verified**. The project applies no parallax correction, so cloud-top positions may be displaced for tall clouds.

13. Storm detection and tracking

Cold-cloud object detection (tobac 1.6.2).

- **Features:** detected on BT at thresholds 273, 253, 235, 221 and 208 K (target = minimum).
- **Minimum sizes:** 25 px at 273 K (100 km²), 8 px at 253 K, 4 px at colder levels. At 4 px the 273 K level produced 3,581 small warm fragments on 14 May, which made trackpy's linking fail.
- **Segmentation:** watershed at 245 K.
- **v1 linking:** trackpy `random` method, `v_max` 30 m/s, 2-frame minimum. The trackpy subnetwork cap was raised to 50 (Phase 4) after 1 May produced a 34-point subnetwork; E8 tracks re-ran byte-identically.
- **v1 merge/split:** tobac `merge_split_MEST` at 25 km.

Tracking audit (Phase 3). Before any cell-level result was used, the v1 tracks were audited on E8. The metrics:

- **direction reversals:** the share of consecutive steps that reverse direction (random = 0.5);
- **ID-switch candidates:** another object overlaps the parent more than the linked one;
- **zero-overlap links;**
- **area jumps:** more than 2× between frames;
- **correlation with the pySTEPS flow.**

Audit metric (E8)	v1 tobac tracks	v2 overlap tracks (adopted)	v2 flow-shifted variant
Cells (≥ 2 h)	613 (181)	88 (23)	95 (22)
Direction reversals (random = 0.5)	0.47 (0.38 cold rows only)	0.35	0.27
ID-switch candidates (fraction of cold links)	0.206	0.004	0.127
Zero-overlap links	0.057	0 (guaranteed by the rule)	0
Area jumps > 2×	0.40	0.28	0.29

Table 4. Tracking audit before and after the fixes. Source: `data/processed/E8_20260514/audit/audit_*.json`; TRACKING_AUDIT.

OBSERVED RESULT: v1 problems.

- 81 % of tracked rows were warm features.
- 15.5 % of steps changed detection level; those steps moved a median 28.5 km, against 19.9 km otherwise.
- Members of the same merge/split family present at the same time were a median 68 km apart (maximum 221 km).
- Nearest-feature truth biased the Phase-2 object results toward persistence.

Fixes (v2, implemented in `ml/tracking/overlap_link.py`).

1. Only cold segments (< 245 K) are treated as storms.
2. **Overlap linking.** A cell continues into the next-frame segment with mutual best overlap. Overlap is measured relative to the smaller object and must be at least 0.2. Extra overlaps are recorded once each as MERGE or SPLIT events, and families are built from them. Linking uses only the current and next frame.
3. **Coldness-weighted centroid.** Pixels are weighted by 245 K – BT, which steadies the position.

Overlap thresholds of 0.1–0.3 gave essentially the same results.

Motion estimation.

- Cell step motion (speed, heading) comes from consecutive centroids. It remains jittery (27–35 % reversals).
- The dashboard's motion arrows therefore use the pySTEPS **field** motion: the cell centre advected 30 min by the Lucas–Kanade field.

On the other events (MULTI_EVENT §6), E10 / E11:

- v1 tracks: reversals 0.46 / 0.46, ID-switch candidates 0.23 / 0.09.
- v2 tracks: reversals 0.33 / 0.21, ID switches 0.00 / 0.00, area jumps 0.28 / 0.24.

LIMITATION.

- Area jumps ($\approx 28\%$) remain. They come from watershed segmentation, not from merges and splits, so areas and growth rates are not reliable.
- There is no independent truth (radar, lightning or manual labels) to decide which linking rule is right, and the PRD's $\geq 90\%$ tracking criterion is not assessed.
- Phase rules v0 are *proposed*, not validated.
- **Object-level claims are diagnostic only, and only to 30 min** (Section 17.9).

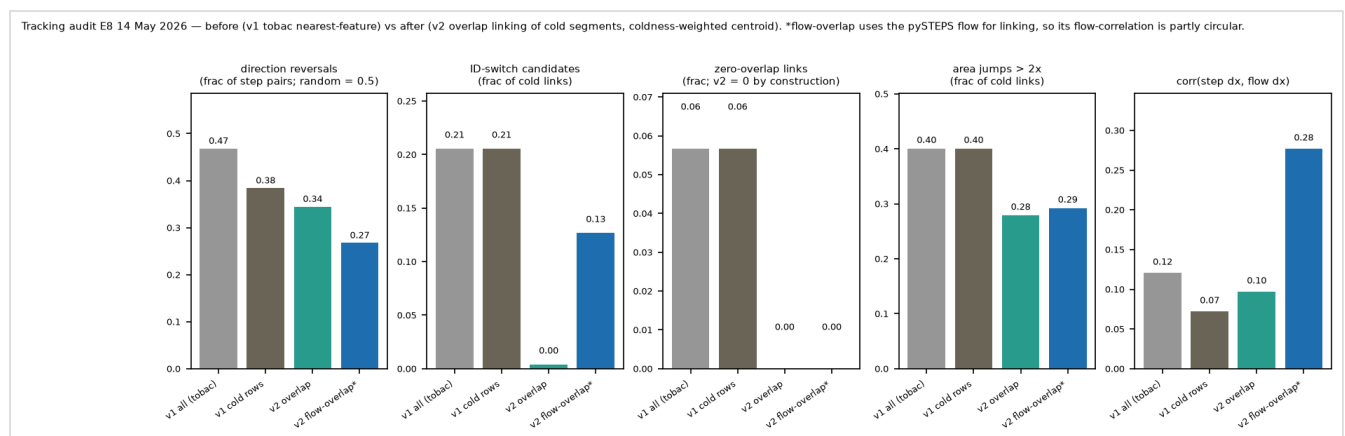


Figure 4. Tracking audit, E8: v1 tobac vs v2 overlap tracks. Source: [data/figures/E8_20260514_fig8_tracking_audit.png](#).

14. Forecasting methods

All forecasts are issued at each 30-min frame from 01:00 to 23:00 UTC (45 issue times per day), for leads of 30, 60 ... 360 min. **A forecast issued at frame k reads only frames $\leq k$** (tested by destroying future frames).

14.1 Persistence

The observed BT field at the issue time is repeated unchanged for every lead: $BT(t + L) = BT(t)$. It has no parameters. It is the minimum reference: any useful method must beat it.

14.2 pySTEPS advection

- **Motion:** pySTEPS 1.21.5 **Lucas–Kanade** dense optical flow, estimated from the last three frames ($t - 60$, $t - 30$, t min). The tracer is cloud-top "coldness", $\max(0, 273 \text{ K} - BT)$, so warm surfaces are ignored.
- **Extrapolation:** semi-Lagrangian advection of $BT(t)$ along that motion for up to 12 steps.
- **Inflow:** pixels whose trajectories come from outside the grid have no forecast (NaN). They are excluded from scoring, and the excluded fraction is reported.

- **Fitting:** none. Settings were fixed in `config/baseline.yaml` before evaluation.
- **Deterministic event:** forecast BT < 235 K.
- **No-fit reference (NP31):** the fraction of pySTEPS event pixels in a 31 × 31-pixel (62 km) window. It gives a probability without any fitting and is used to judge whether the ML adds more than smoothing.

14.3 ML prototype

A per-pixel probability that $BT(t + L) < 235$ K, from a histogram gradient-boosting classifier with isotonic calibration. It uses the pySTEPS forecast and recent-IR features (Section 15). The deterministic ML event is $p \geq 0.5$, the threshold pre-declared from VALIDATION_PLAN §2.

Decay-aware extension: not built (Phase 5).

- The simplest extension would extrapolate each cell's recent Lagrangian tendency (min-BT or log-area change), with AR(1) damping.
- A rule set before computing was: build only if a tendency persists on the development event (E8), with lag-1 coefficient $\phi > 0$ and a 95 % cell-bootstrap interval excluding 0.
- No tendency passed. Min-BT $\phi = -0.03$ [−0.22, 0.15]; log-area $\phi = 0.16$ [−0.01, 0.31]; cold-core $\phi = -0.25$ [−0.53, 0.03]. **Decision: STOP.**

15. ML model

Item	Value (frozen)	Source
Model version	ml-v0	<code>data/models/ml-v0/model_card.json</code>
Model type	<code>sklearn.ensemble.HistGradientBoostingClassifier</code> (histogram GBDT, the LightGBM family; LightGBM not installed) + <code>IsotonicRegression</code>	<code>config/ml_dataset.yaml</code>
Hyper-parameters (fixed before training)	learning rate 0.05; max 31 leaf nodes; min 500 samples per leaf; L2 1.0; no internal early stopping (it would split pixels at random); random_state 0	model card
Number of trees	216 , chosen by minimum weighted validation log-loss over 400 staged iterations, then refit	model card; fig. 21
Target	Observed $BT(t + L) < 235$ K at each 2 km pixel	<code>config/ml_dataset.yaml</code>
Lead times	30, 60, 90 ... 360 min (12 leads)	—
Features (17)	lead; pySTEPS BT at $t + L$; pySTEPS event fraction in 11 × 11 and 31 × 31 px; pySTEPS min in 11 × 11 px; pySTEPS valid fraction in 31 × 31 px (edge proxy); BT(t); BT(t) event fraction 31 × 31; BT(t) min 11 × 11; 30-min tendency (raw and advected); neighbourhood mean of advected tendency; motion speed; domain cold fraction and its 60-min change; valid hour (sin, cos). No latitude, longitude or terrain.	<code>ml/features/grid_features.py</code>
Training period	18–31 May 2026: 14 days, of which 7 convective; 1,467,318 sampled rows	model card
Calibration / validation period	6–12 May 2026: 7 days, of which 5 convective; 856,830 rows; used only for tree count and isotonic calibration	model card
Test period (held out)	E8 14 May, E10 4 May, E11 16 May 2026; never read in training; scored once with the frozen model	ML_PROTOTYPE §1
Excluded	All event days (1, 4, 14, 16 May) ± 1 day	dataset manifest
Sampling	Per (day, issue, lead): 150 event, 150 near-cold non-event, 60 far non-event pixels, with inverse-probability weights (weighted totals reproduce the full grid, tested)	<code>config/ml_dataset.yaml</code>
Sufficiency gate	≥ 7 convective training days and ≥ 3 convective validation days; result 7 and 5, so PROCEED	model card
Calibration	Isotonic per lead bin (30 / 60 / 90–120 / 150–360 min), validation days only. Weighted log-loss 0.0552 raw → 0.0529 calibrated	model card

Item	Value (frozen)	Source
Deterministic threshold	$p \geq 0.5$ (pre-declared, not tuned)	VALIDATION_PLAN §2
Freeze	model.joblib SHA-256 597e3c06c5db7ec2...; calibrators.joblib 0da01eb8bd2f3598...; model_card.json 2a763b0aaeae9858...	docs/FREEZE_ml-v0.js on; PROVENANCE §Phase 7

Table 5. ML model card summary.

INTERPRETATION: what the model learned.

- Permutation importance on a validation-day subsample ranks the features by the increase in weighted log-loss:

Feature	Log-loss increase
pySTEPS 31-px event fraction	0.019
Lead	0.008
BT(t) neighbourhood fraction	0.005
Motion speed	0.003
pySTEPS minimum	0.003
Advection-tendency mean	0.003

- Diurnal terms add ≈ 0.001 – 0.002 . Domain-scale and raw-BT features add nothing (≤ 0).
- **The model is mainly a calibrated, lead-aware neighbourhood refinement of pySTEPS, with modest tendency and time-of-day information.**
- **Storm initiation (new development) was not demonstrated**, and no initiation-specific score exists.
- After calibration, probabilities from 150 min onward never reach 0.5, because high raw probabilities verified only about 27 % of the time on validation days.

16. Validation methodology

Event selection. Rules fixed before any scoring (`config/events.yaml`):

- The date is printed in the archived IMD hail report.
- All 24 hourly CPC files exist.
- At least 2 reported places geocode inside the domain.
- The date is not within ± 1 day of another selected event.

Suitability rule (also fixed beforehand): all 48 frames present, and BT < 235 K present in ≥ 50 % of frames.

- **Results:** E8, E10 and E11 passed. E8a (1 May) failed with 21 % deep-convection frames and was excluded.
- **Unavailable:** E9 (5 and 9 April 2026), because April is missing from the archive.

Time-blocked separation.

- **Grid baselines:** no fitted parameters, so the only leakage risk is future data; every forecast uses frames $\leq$ issue time (tested).
- **ML:** calendar-day blocks, no random pixel splits:
 - train 18–31 May;
 - validation 6–12 May;
 - test E8/E10/E11;
 - ± 1 -day buffers around every event day.
- **LIMITATION:** the archive starts 1 May, so training days come **after** the test days (backward-in-time test).

Leakage checks (automated tests):

- features unchanged when future frames are destroyed;

- no test or buffer day in any training or validation file;
- lifecycle tendencies unchanged when later rows are deleted;
- isotonic calibration reproduces its validation log-loss (it was fitted on validation rows);
- a split touching a test day is rejected.

Identical-pixel scoring. Mask **V** = inside the domain, and valid in the observation, the persistence forecast and the pySTEPS forecast. Every method is scored on the same pixels (tested). A second mask **D** includes the inflow edge, where pySTEPS has no forecast; D is used for edge-only diagnostics.

Metrics (VALIDATION_PLAN §2), from the contingency table of hits H, misses M and false alarms F:

Metric	Definition	What it measures
POD	$H/(H+M)$	Share of observed events that were forecast
FAR	$F/(H+F)$	Share of forecast events that did not happen
CSI	$H/(H+M+F)$	Overall yes/no agreement
Frequency bias	$(H+F)/(H+M)$	Over- (> 1) or under-forecasting (< 1) of event area
FSS	Roberts & Lean (2008), at 10 / 20 / 40 km	Spatial agreement at a given scale; "useful" if $\geq 0.5 + f_0/2$
BSS	$1 - BS/BS_{ref}$, reference = training-split climatology per lead	Probability skill; deterministic methods enter as 0/1 probabilities
Reliability / ECE	10 bins; $ECE = \sum \sum p - \sum o /n$	Calibration

Uncertainty.

- Within each event: a circular block bootstrap over issue times, with 6-h blocks (the longest lead), 1000 resamples, seed 20260925, and hits/misses/false alarms (or Brier sums) re-pooled on each resample.
- Across events: a bootstrap by event day. With 3 events it is **indicative only**; the best possible one-sided sign-test p is 0.125.

Error decomposition (Phase 5). Pooled frequency bias factorises exactly as $\text{bias} = \mathbf{E} \times \mathbf{C} \times \mathbf{G}$:

- **E (edge):** the effect of scoring only the region pySTEPS can forecast;
- **C:** the change in advected cold area;
- **G:** the observed domain cold-area change, i.e. unchanged intensity.

An oracle that matches forecast area to the observation bounds what any uniform growth/decay correction could gain. It uses the observation and is not a forecast.

Reproducibility. Reruns produced byte-identical CSVs (baselines, v2 tracks, cross-event tables, dataset days, reliability gate). Every output is listed with its SHA-256 in per-phase provenance files, and all 386 evidence files are frozen (Section 22).

17. Results

All values: held-out event days, mask V, event = BT < 235 K.

- P = persistence
- A = pySTEPS advection
- NP31 = no-fit neighbourhood pySTEPS
- ML = ml-v0 calibrated

17.1 Single-event baseline (E8, 14 May 2026)

Lead (min)	Excluded fraction	CSI P / A	POD P / A	FAR P / A	Bias P / A	FSS 20 km P / A	FSS 40 km P / A
30	0.087	0.509 / 0.593	0.688 / 0.802	0.338 / 0.306	1.04 / 1.16	0.792 / 0.853	0.851 / 0.898
60	0.112	0.352 / 0.428	0.534 / 0.684	0.493 / 0.467	1.05 / 1.28	0.635 / 0.713	0.700 / 0.771
120	0.162	0.191 / 0.245	0.327 / 0.481	0.684 / 0.666	1.03 / 1.44	0.400 / 0.480	0.448 / 0.534
180	0.211	0.106 / 0.156	0.192 / 0.351	0.809 / 0.780	1.01 / 1.60	0.237 / 0.333	0.266 / 0.371
240	0.258	0.077 / 0.109	0.139 / 0.267	0.852 / 0.844	0.94 / 1.71	0.169 / 0.244	0.187 / 0.271
360	0.348	0.038 / 0.043	0.066 / 0.148	0.919 / 0.943	0.82 / 2.61	0.084 / 0.099	0.097 / 0.110

Table 6. E8 grid baselines, pooled over 45 issue times. Source: data/processed/E8_20260514/baseline/grid_metrics_by_lead.csv; BASELINE_EVAL §3. The useful FSS threshold is ≈ 0.51 .

17.2 Baselines on three events (Phase 4)

Lead (min)	CSI P / A: E8	CSI P / A: E10	CSI P / A: E11	Bias P / A: E8	Bias P / A: E10	Bias P / A: E11
30	0.509 / 0.593	0.514 / 0.582	0.492 / 0.589	1.04 / 1.16	1.08 / 1.11	1.02 / 1.13
60	0.352 / 0.428	0.336 / 0.428	0.341 / 0.441	1.05 / 1.28	1.13 / 1.18	1.01 / 1.23
120	0.191 / 0.245	0.169 / 0.263	0.160 / 0.283	1.03 / 1.44	1.11 / 1.21	0.85 / 1.40
180	0.106 / 0.156	0.089 / 0.178	0.025 / 0.213	1.01 / 1.60	1.04 / 1.15	0.56 / 1.56
240	0.077 / 0.109	0.043 / 0.130	0.001 / 0.110	0.94 / 1.71	0.98 / 1.07	0.41 / 1.68
360	0.038 / 0.043	0.019 / 0.054	0.000 / 0.002	0.82 / 2.61	0.91 / 0.97	0.80 / 1.53

Table 7. Per-event CSI and frequency bias. Source: data/processed/multi_event/event_metrics_by_lead.csv; MULTI_EVENT §4.

Lead (min)	Pooled CSI P	Pooled CSI A	Events with A better (CSI and FSS 40 km)
30	0.511	0.586	3 / 3
60	0.342	0.429	3 / 3
120	0.176	0.258	3 / 3
180	0.092	0.172	3 / 3
360	0.022	0.049	3 / 3

Table 8. Cross-event pooled CSI. Source: data/processed/multi_event/cross_event_by_lead.csv.

	E8	E10	E11
Useful FSS 40 km (P / A)	to 90 / 120 min	to 90 / 120 min	to 90 / 120 min
Useful FSS 20 km (P / A)	to 60 / 90 min	to 60 / 90 min	to 60 / 120 min
CSI e-folding time, 30–180 min fit (P / A)	96 / 113 min	87 / 128 min	54 / 149 min
Within-event 95 % interval of CSI(A) – CSI(P) excludes 0 at	30–240 min	30–360 min	30 min only

Table 9. Skill decay and within-event uncertainty. Source: decay_by_event.csv, event_block_bootstrap.csv; MULTI_EVENT §4–5.

- **OBSERVED RESULT.** pySTEPS beats persistence on CSI and FSS 40 km at every lead on all 3 days. Useful skill ends at about 2 h.
- **INTERPRETATION.** The Phase-2 finding on E8 was repeated on two more days.
- **LIMITATION.** Three events cannot give cross-event significance (best sign test $p = 0.125$). E11 is weak (base rate 0.4 %), so its intervals are wide.

Phase 4: persistence (grey) vs pySTEPS advection (blue) on each event — identical protocol (config/baseline.yaml unchanged), 30-min leads, same scored pixels for both methods. Data: NOAA/NCEP/CPC merged IR.

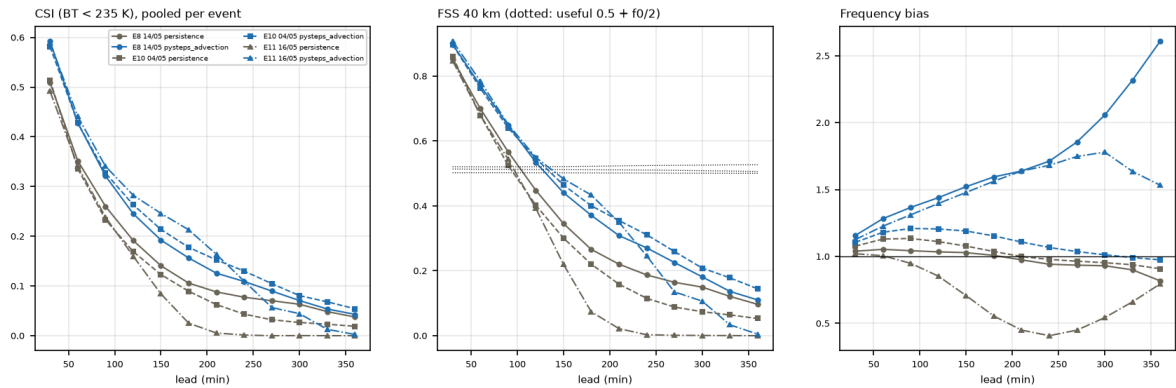


Figure 5. CSI, FSS 40 km and frequency bias vs lead for each event. Source: [data/figures/multi_event_fig11_skill_by_event.png](#).

17.3 Why skill fades: bias decomposition (Phase 5)

Event	Lead (min)	Bias (scored region)	Edge E	Advection area C	Intensity G	Whole-domain bias	Observed cold pixels in edge	CSI P / A / oracle
E8	120	1.44	1.33	1.08	1.00	1.09	30 %	0.191 / 0.245 / 0.269
E8	360	2.61	3.76	0.73	0.96	0.70	76 %	0.038 / 0.043 / 0.048
E10	120	1.21	1.11	1.03	1.06	1.09	17 %	0.169 / 0.263 / 0.283
E11	120	1.40	1.19	1.22	0.97	1.18	19 %	0.160 / 0.283 / 0.337
E11	360	1.53	4.80	1.25	0.26	0.32	80 %	0.000 / 0.002 / 0.010

Table 10. Exact bias factorisation (bias = $E \times C \times G$). Source: [data/processed/multi_event/phase5/decomposition_by_lead.csv](#); DECAY_AWARE §2.

- **OBSERVED RESULT.**
 - The growing pySTEPS bias is mostly the edge term. On E8 at 6 h, the bias is 2.61 on the scored region but 0.70 over the whole domain.
 - Pooled over each day, the intensity term is small on E8/E10.
 - By time of day it shows growth for morning issues and decay for afternoon/evening issues.
- **INTERPRETATION.**
 - New and entering storms upstream cannot be forecast by extrapolation.
 - Even a perfect uniform area correction adds at most +0.035 CSI on E8 and +0.026 on E10.
 - Placement error and new development dominate the remaining error.
- **LIMITATION.** The oracle covers uniform corrections only.

Phase 5 task 1 — where the pySTEPS frequency-bias growth comes from. $E > 1$: observed events sit disproportionately in the unscorable inflow edge; G: observed domain cold-area change ignored by extrapolation. Pooled over issue times.

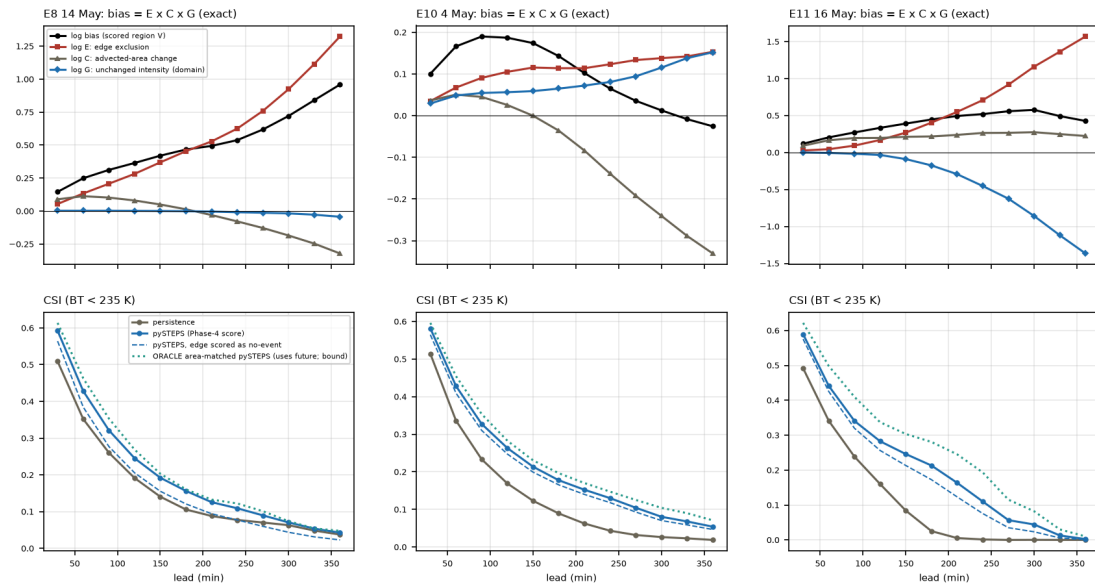


Figure 6. Bias decomposition and CSI (persistence, pySTEPS, edge-scored, oracle) by lead. Source: data/figures/phase5_fig14_bias_decomposition.png.

17.4 ML vs baselines (Phase 6, held-out E8 + E10 + E11 pooled)

Lead (min)	CSI P / A / NP31 / ML ($p \geq 0.5$)	FSS 40 km A / ML (event mean)	Bias A / ML	BSS P / A / NP31 / ML
30	0.511 / 0.586 / 0.596 / 0.583	0.901 / 0.886	1.13 / 0.76	0.32 / 0.44 / 0.63 / 0.67
60	0.342 / 0.429 / 0.444 / 0.452	0.773 / 0.778	1.22 / 0.66	-0.04 / 0.10 / 0.41 / 0.51
120	0.176 / 0.258 / 0.264 / 0.243	0.543 / 0.525	1.30 / 0.54	-0.47 / -0.38 / 0.02 / 0.27
180	0.092 / 0.172 / 0.172 / 0.000	0.402 / 0.000	1.32 / 0.00	-0.69 / -0.67 / -0.23 / 0.13
240	0.052 / 0.122 / 0.116 / 0.000	0.276 / 0.000	1.28 / 0.00	-0.78 / -0.82 / -0.37 / 0.06
360	0.022 / 0.049 / 0.045 / 0.000	0.086 / 0.000	1.29 / 0.00	-0.84 / -1.11 / -0.59 / -0.03

Table 11. Final comparison. Source: data/processed/ml_eval/ml-v0/metrics_pooled_lead.csv, metrics_by_event_lead.csv; ML_PROTOTYPE §3; FINAL_RESULTS Slide 1.

Event	30 min	60 min	120 min	180 min
E8	+0.04 [0.03, 0.11]	+0.12 [0.04, 0.36]	+0.27 [0.04, 0.92]	+0.49 [0.10, 1.63]
E10	+0.03 [0.01, 0.11]	+0.11 [0.03, 0.50]	+0.26 [0.08, 0.95]	+0.31 [0.05, 1.37]
E11	0.00 [-0.03, 0.09]	+0.01 [-0.07, 0.19]	+0.16 [-0.12, 0.86]	+0.26 [-0.17, 1.46]

Table 12. BSS(ML) – BSS(NP31) per event, with 95 % within-event block-bootstrap intervals. Source: bootstrap_ml_vs_pysteps.csv; ML_PROTOTYPE §3.

• OBSERVED RESULT.

- ML has the highest BSS at every lead to 4 h (0.67, 0.51, 0.27, 0.13, 0.06 at 30 min to 4 h), and it is -0.03 at 6 h.
- Its gain over the no-fit neighbourhood reference is significant within E8 and E10, but not within E11.
- As a yes/no forecast ($p \geq 0.5$), ML CSI beats pySTEPS only at 60 min (0.452 vs 0.429). It is lower at 30 and 120 min, and 0 from 150 min.
- The CSI(ML) – CSI(pySTEPS) intervals include 0 at 30–120 min on all events. From 150 min the upper bound is ≤ 0 .

• INTERPRETATION.

- The ML advantage is in **probability skill**, not in yes/no maps beyond 1 h.

- The calibrated model will not issue high probabilities at long leads, which is honest calibration rather than a display failure.
- **LIMITATION.**
 - This is not "ML beats pySTEPS" in general.
 - There is no 6-h skill, and no skill across events can be established from 3 days.
 - The BSS reference (training climatology) differs from the test-day base rates.

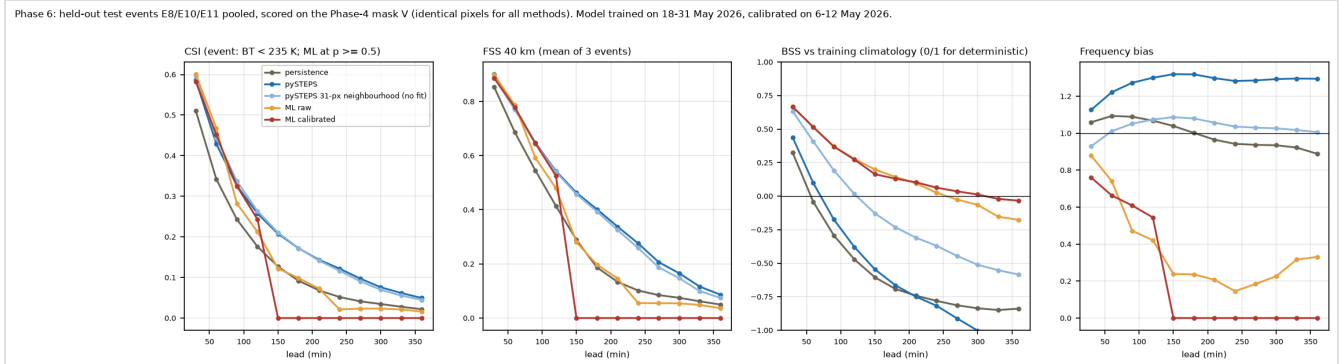


Figure 7. CSI, FSS 40 km, BSS and bias vs lead for all methods, held-out events pooled. Source: `data/figures/phase6_fig17_skill_vs_lead.png`.

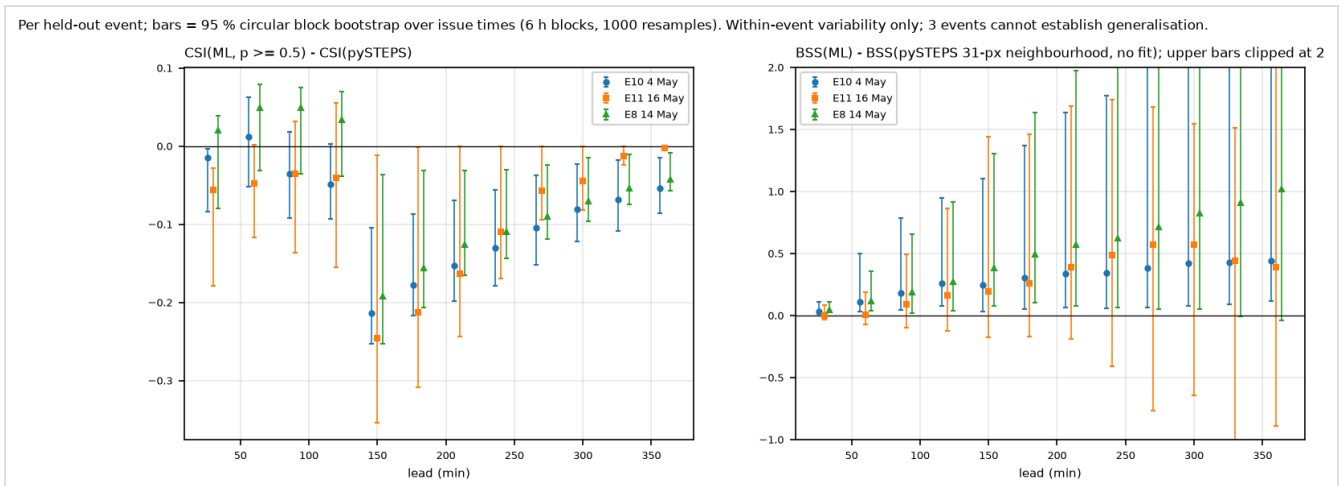


Figure 8. Per-event CSI(ML) – CSI(pySTEPS) and BSS(ML) – BSS(NP31), with block-bootstrap intervals. Source: `data/figures/phase6_fig18_ml_minus_pysteps_ci.png`.

17.5 Calibration

Lead bin (min)	ECE (ML)	ECE (NP31)
30	0.0022	0.0023
60	0.0034	0.0082
90–120	0.0042	0.0163
150–360	0.0026	0.0301

Table 13. Expected calibration error on held-out days (10 bins). Source: `data/processed/ml_eval/ml-v0/reliability.csv`; FINAL_CLAIMS A7.

- **OBSERVED RESULT.** ML probabilities are close to the reliability diagonal in every lead bin. NP31 is overconfident from 60 min on.

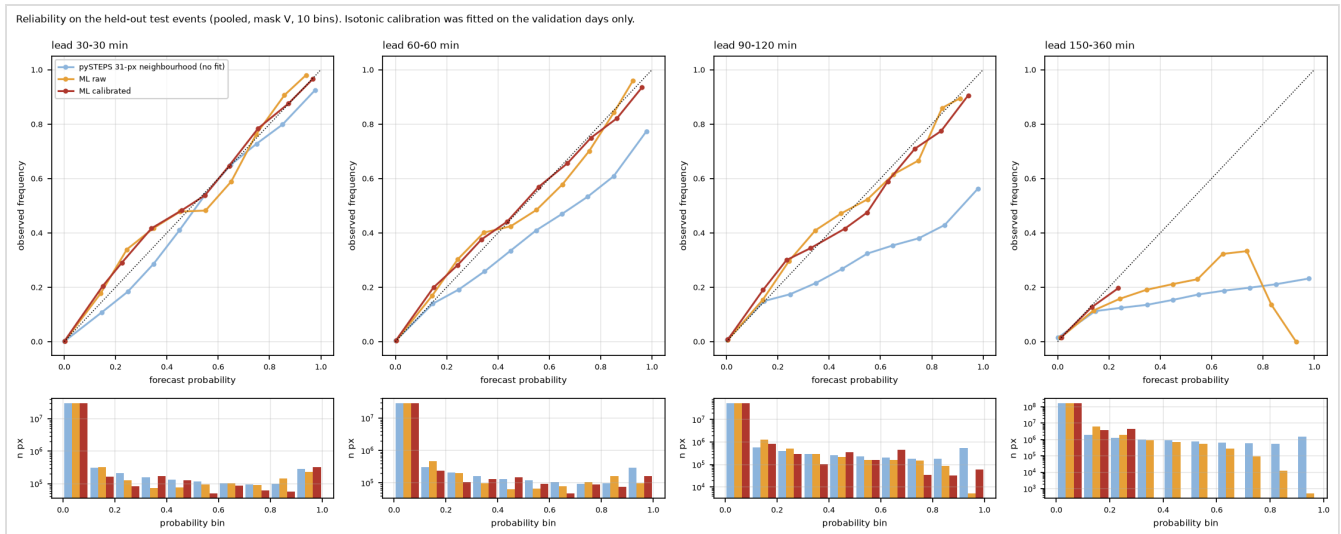


Figure 9. Reliability diagrams with counts, per lead bin. Source: data/figures/phase6_fig19_reliability.png.

17.6 Inflow edge

- **OBSERVED RESULT.** On pixels that pySTEPS cannot forecast (outside V):
 - ML BSS is 0.43 at 30 min and 0.22 at 120 min, vs 0.37 and 0.11 for NP31;
 - at 30 min, ML POD is 0.38 vs 0.21 for pySTEPS.
- **INTERPRETATION.** This is the one place the model adds placement information for entering storms. Source: metrics_edge_only.csv.

17.7 Event-wise findings

Event	Main finding	Caveat
E8 (14 May)	Strong day (base rate 2.7 %); pySTEPS > P at all leads; ML > NP31 (intervals above 0, 30 min to 3 h)	6 QC-fail frames (10:00–12:30 UTC)
E10 (4 May)	Strongest day (base rate 4.2 %); smallest pySTEPS bias growth (1.11 → 1.21, then ≈ 1.0)	9 QC-fail frames (17:00–21:30 UTC)
E11 (16 May)	Weak day (base rate 0.4 %); pySTEPS > P; ML not significantly better than NP31	Small sample; wide intervals

Table 14. Event-wise summary. Sources: MULTI_EVENT §4–7; ML_PROTOTYPE §3.

17.8 Tracking audit results

See Table 4 (Section 13). **OBSERVED RESULT:** v2 removes ID switches (0.206 → 0.004 on E8; 0.00 on E10 and E11). **LIMITATION:** area jumps (≈ 0.28) and 21–35 % direction reversals remain.

17.9 Object diagnostics (diagnostic only)

- **OBSERVED RESULT.** At 30 min, pySTEPS-field advection places cold cells closer than persistence on all 3 events: 0.59 / 0.62 / 0.57 of paired cases (43 / 66 / 18 cells).
- **LIMITATION.**
 - At 60 min the advantage fails on E11 (0.42, 15 cells), so object claims stop at 30 min.
 - Cell positions have no independent truth.
- Source: object_diagnostics_by_event.csv.

17.10 Reproducibility and test results

See Section 22. **OBSERVED RESULT:** 76 Python tests pass (1 skipped), 6/6 web unit tests, 2/2 browser tests; the build passes; the freeze guard confirms that all 386 frozen files are unchanged.

18. Real event validation

Code	Date	Reported hail places (in domain)	Frames	Issue times	QC-fail frames	Deep-convection frames	Role
E8	14 May 2026	16 (15)	48	45	6	92 %	Phase-1 reference; held-out test
E10	4 May 2026	10 (10)	48	45	9	96 %	Held-out test
E11	16 May 2026	5 (3)	48	45	0	65 %	Held-out test (weak event)
E8a	1 May 2026	20 (17)	48	—	0	21 %	Excluded: fails the pre-declared suitability rule (≥ 50 %)
E9	21 Mar, 5 and 9 Apr 2026	—	0	—	—	—	Unavailable: before 30 Mar / April missing from the archive
E1–E7	2018–2025	—	0	—	—	—	Not available: before the approved archive (or outside the domain)

Table 15. Event table. Sources: MULTI_EVENT §1, event_qc.csv, REPLAY_SPEC §1.

Selection rationale. The dates come from the archived IMD RMC New Delhi hail report. Candidates were ranked by number of reported places and filtered by the rule in Section 16. No other candidate qualified: 13, 15, 2, 3 and 5 May are within ± 1 day of a selected event; 6 May has 1 in-domain place; 7 May has 1 place; and the PDF ends on 16 May.

Held-out evaluation. All three days are excluded, with ± 1 -day buffers, from ML training and calibration. The grid baselines have no fitted parameters.

LIMITATION: scope. Three pre-monsoon days in May 2026, one region (NW India) and one hail report. Nothing is shown for the monsoon, cloudbursts, other regions, other years or other satellites. Hail reports are used only to choose days; they were never verification truth.

19. Product / dashboard

The **web prototype** (`apps/web/`) is a static React 18 + TypeScript + Vite + MapLibre application.

- **Data:** it reads replay bundles (`apps/web/public/bundles/{event}/`) built by `pipelines/replay/build_bundle.py` from frozen files only; each input hash is checked against the freeze manifest.
- **Bundles:** one per held-out event (E8, E10, E11), each with 20 frozen inputs and 735 output files; 82 MB in total.
- **Tiles:** fields are stored as 8-bit data tiles on a Web-Mercator display grid and colourised in the browser. The display resampling (nearest neighbour) is for drawing only.
- **App shell** (UI_UX_SPEC §3):
 - sidebar with START / OPERATE / INVESTIGATE / EVIDENCE groups and count badges;
 - top bar with the REPLAY badge, valid time in UTC and IST, data tier, model version and frame QC;
 - a 48-frame timeline with QC-fail marks, play at 1×/2×/5×, and a lead selector;
 - keyboard shortcuts (←/→, Space, 1–9, B);
 - the footer disclaimer "*Research prototype — not an official IMD warning*".

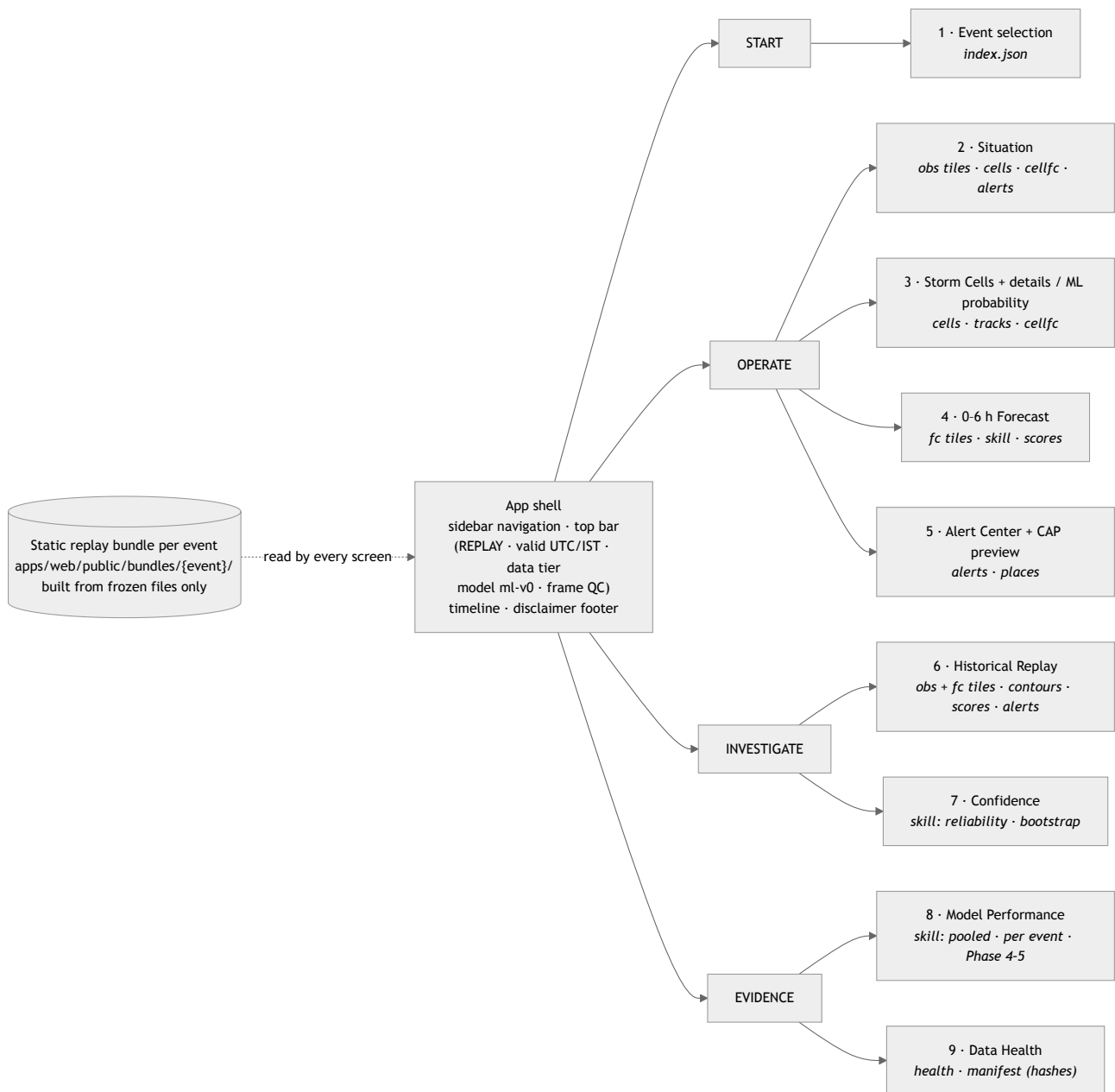


Figure 10a (Diagram 5). Dashboard information architecture: navigation groups, screens and the bundle files each screen reads.

#	Screen	Purpose	Inputs (bundle)	Displayed information	Decision-support value	Status
1	Event selection	Choose a real event	index.json	Cards E1–E11, with BUNDLED / EXCLUDED / NOT AVAILABLE and the reason	Shows which evidence exists and why others do not	Implemented
2	Situation dashboard	Current overview	obs tiles, cells, cellfc, alerts	IR map, phase-styled cells (colour + line style), pySTEPS motion arrows; KPIs (active cells, Mature cells, places with $P \geq 0.5$ in 60 min, frame QC); top-5 cells by ML risk; source status	Where deep convection is now, and which cells to watch	Implemented
3	Storm Cell Tracker	Inspect one storm	cells, tracks, cellfc	Selected cell outline; track so far (solid); pySTEPS-advected track to +6 h (dashed, labelled); sortable cell table	Follow a cell's history and likely path	Implemented

#	Screen	Purpose	Inputs (bundle)	Displayed information	Decision-support value	Status
4	0–6 h Forecast	Forecast at a lead	fc tiles, skill, scores	Persistence / pySTEPS / ML probability at T+30 min to T+6 h; probability threshold; optional observed outcome (replay); skill ribbon (validated CSI, BSS, FSS 40 km, bias; de-emphasised beyond validated skill); envelope; this issue's scores	The forecast with its trustworthiness shown next to it	Implemented (T+15 min disabled: 30-min source)
5	Storm details / ML probability	Cell-level probability	cells, cellfc	Cell card: phase (rules v0, PROPOSED), min BT, ΔBT 30 min, area, cold core, age so far, merges/splits, motion; sparklines; ML probability by lead near the forecast cell position	Quick risk read for a single storm	Implemented (display summary, not a verified cell metric)
6	Historical Replay	Honest look-back	obs + fc tiles, contours, scores, alerts	Three synced panes: KNEW / PREDICTED / HAPPENED; method toggle (B); lead Δ; data-availability row; per-issue scorecard; would-have-issued drafts	Lets evaluators see what the system would have said, and how it turned out	Implemented
7	Model Performance	Evidence	skill	BSS and CSI vs lead (pooled or per event); final comparison table; Phase-4 consistency; Phase-5 decomposition; claim-guard banner	Shows where each method is and is not useful	Implemented (no client-side computation)
8	Confidence / Data Health	Trust and provenance	skill, health, manifest	Reliability diagrams + ECE; per-event intervals. Source status; per-frame QC strip with raw-file SHA-256; unscorable fraction by lead; model card; limitations; bundle input hashes	Shows when to distrust the output	Implemented
9	Alert Center	Human-in-the-loop advisories	alerts, places	System status; draft advisories (ML $P \geq 0.5$ within 10 km of a report place at +30/+60 min); Approve / Reject; audit log	Nothing issued without approval	Implemented (exercise only; alert skill not verified)
10	Exercise CAP preview	Standard message	approved draft	CAP 1.2 XML preview and download: status=Exercise, disclaimer, source prediction ID, 10 km circle	Interoperable format for a future workflow	Implemented (exercise only)

Table 16. Dashboard screens. Sources: PRODUCT_DEMO; `apps/web/src/screens/*`.

Screenshot provenance. Figures 10–14, 16 and 17 are screenshots of the production build replaying the frozen E8 bundle, taken during the verified local browser run. The files are in `apps/web/e2e-shots/`, which is git-ignored. Basemap-off versions are regenerated by `npm run e2e`.

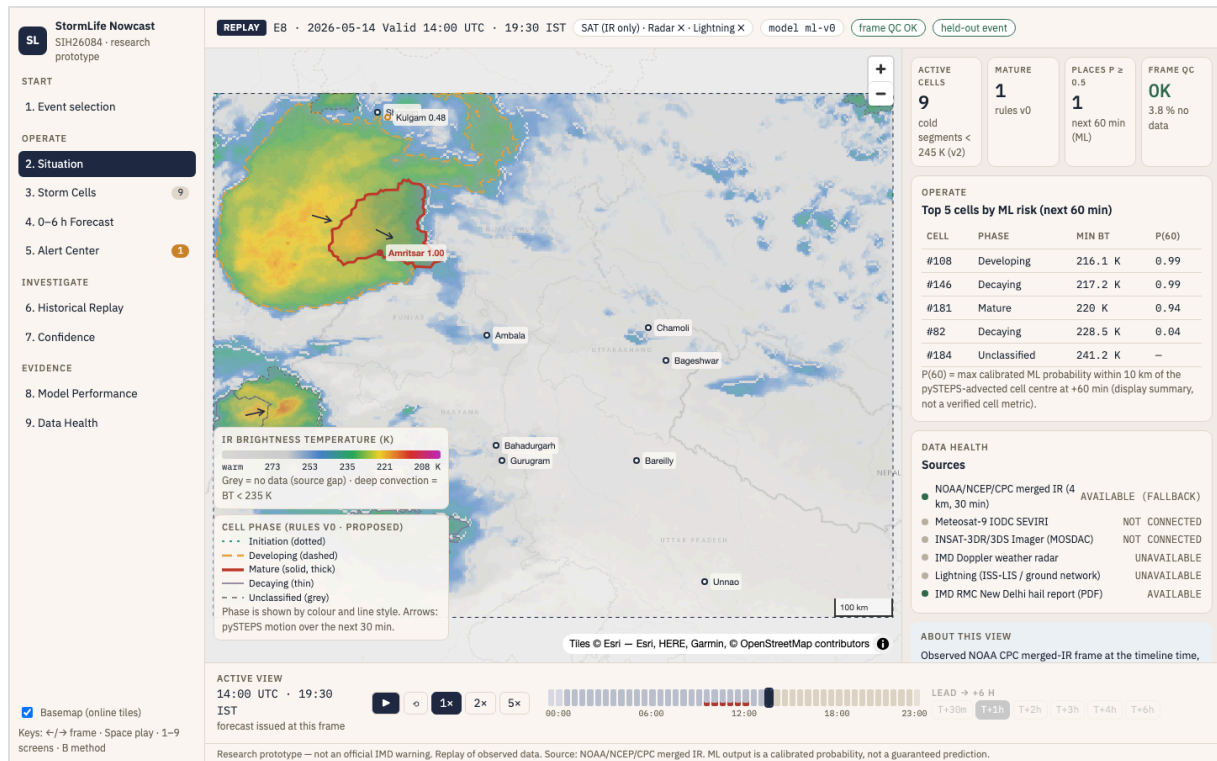


Figure 10. Situation dashboard, E8 14 May 2026, 14:00 UTC: 9 active cells; Amritsar flagged by the ML probability; source status. Screenshot from the verified local browser run.

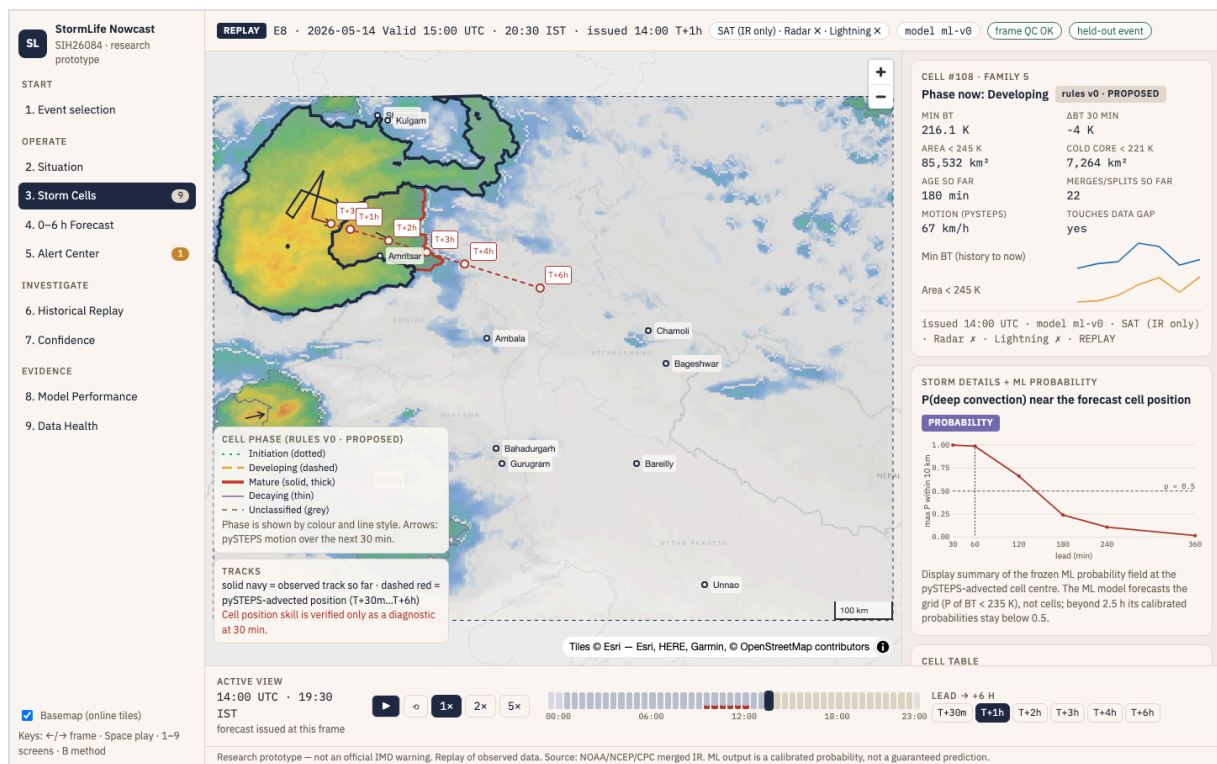


Figure 11. Storm Cell Tracker: cell #108 (Developing, min BT 216.1 K), selected by clicking the map, with its observed track, pySTEPS-advected track and ML probability by lead.

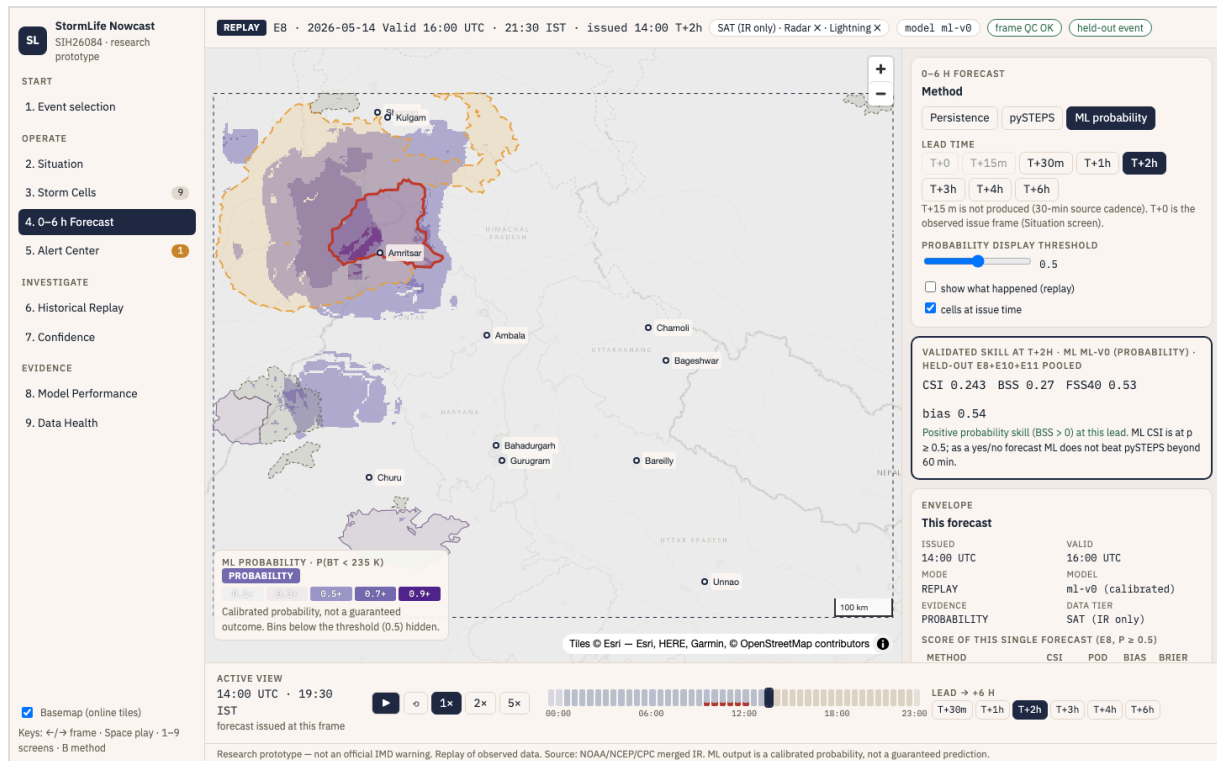


Figure 12. 0–6 h Forecast, ML probability at T+2h (display threshold 0.5). Skill ribbon: CSI 0.243 · BSS 0.27 · FSS40 0.53 · bias 0.54, with the yes/no caveat.

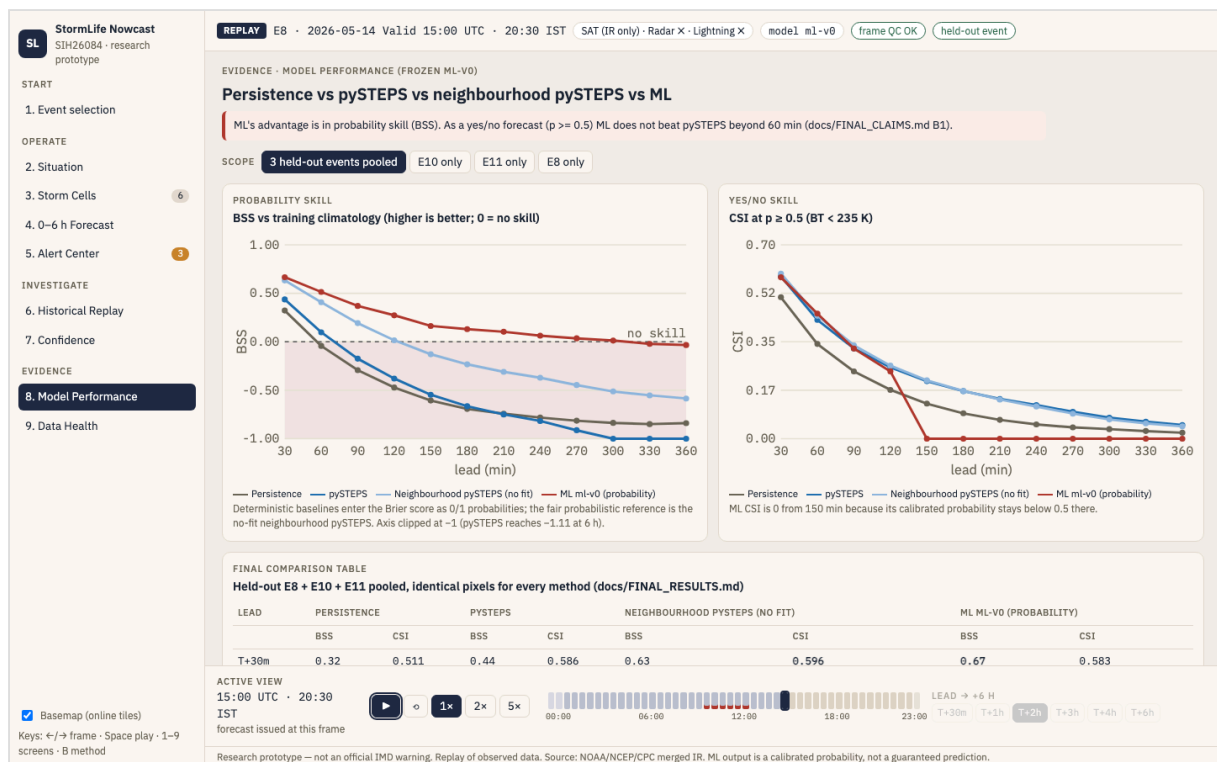


Figure 13. Model Performance screen: claim-guard banner, BSS and CSI vs lead, final comparison table.

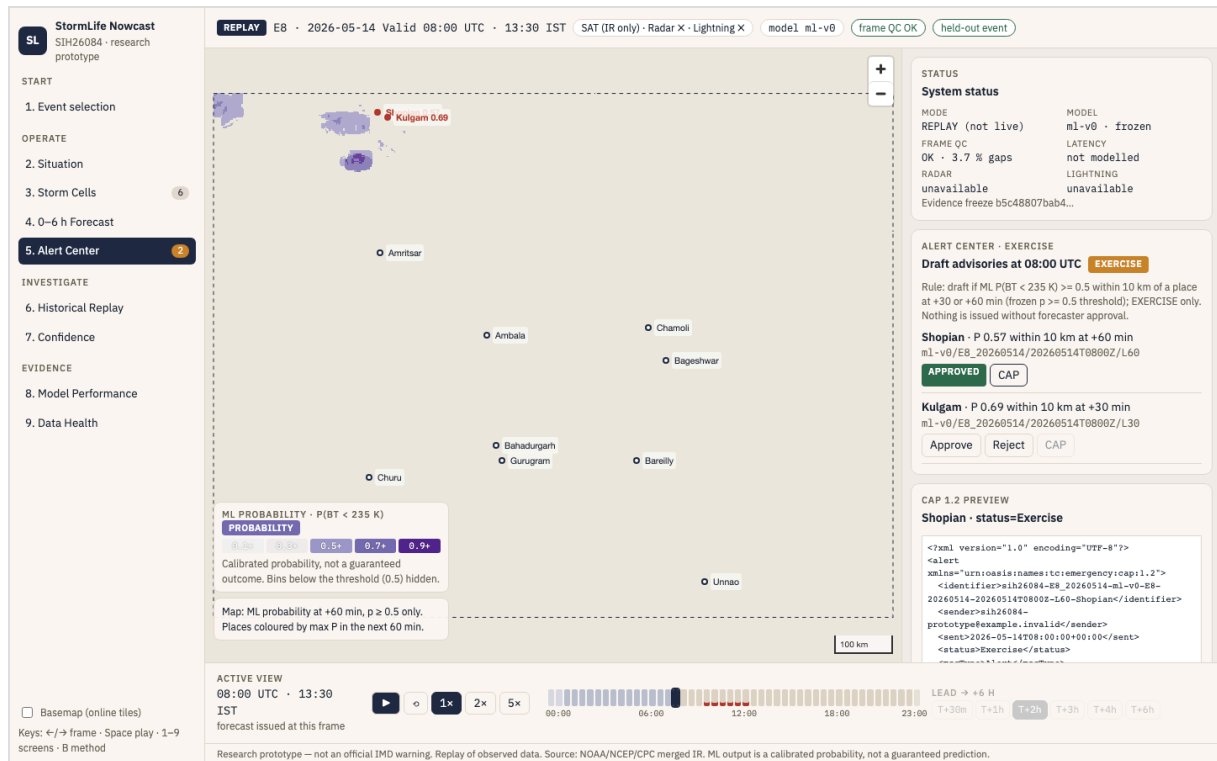


Figure 14. Alert Center at 08:00 UTC: exercise draft advisories (Shopian P 0.57 at +60 min, Kulgam P 0.69 at +30 min), one approved, with the CAP 1.2 preview (status=Exercise).

20. Historical replay

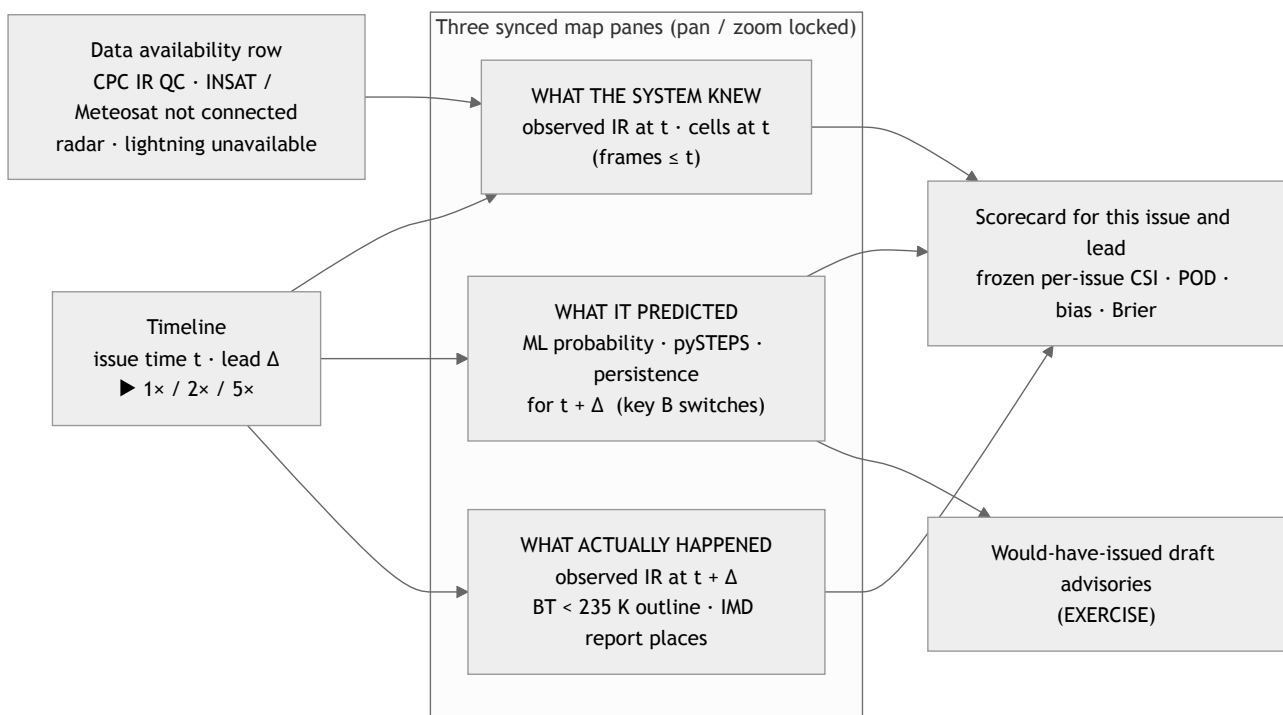


Figure 15 (Diagram 4). Historical replay workflow.

The replay principle (REPLAY_SPEC). At issue time t the screen shows three synced maps with pan and zoom locked:

- **What the system knew:** the observed IR frame at t and the cells at t . Only frames up to t are used.
- **What it predicted:** the forecast for $t + \Delta$. Key **B** cycles ML probability $\rightarrow$ pySTEPS $\rightarrow$ persistence, so comparisons are like-for-like.
- **What actually happened:** the observed IR at $t + \Delta$, the observed BT < 235 K outline, and the IMD-report places.

Timeline.

- The timeline steps through the 48 frames. Forecasts exist for the 45 issue times from 01:00 to 23:00 UTC.
- $\Delta \in \{30, 60, 120, 180, 240, 360\}$ min.
- Playback runs at $1\times$ (one 30-min step per 2 s), $2\times$ or $5\times$.

Below the maps:

- the per-issue scorecard, which shows the frozen scores of all four methods at that lead;
- the would-have-issued exercise drafts;
- a data-availability row: CPC IR QC status; INSAT not available operationally; Meteosat not connected; radar and lightning unavailable.

LIMITATION.

- This is a **replay of saved outputs**, not a live feed. Product latency is not modelled.
- Hail reports have dates only, so alert lead time relative to reported impact cannot be computed.
- One minor exception to "history only": cells seen in a single frame were dropped offline using the next frame. This is stated on screen.

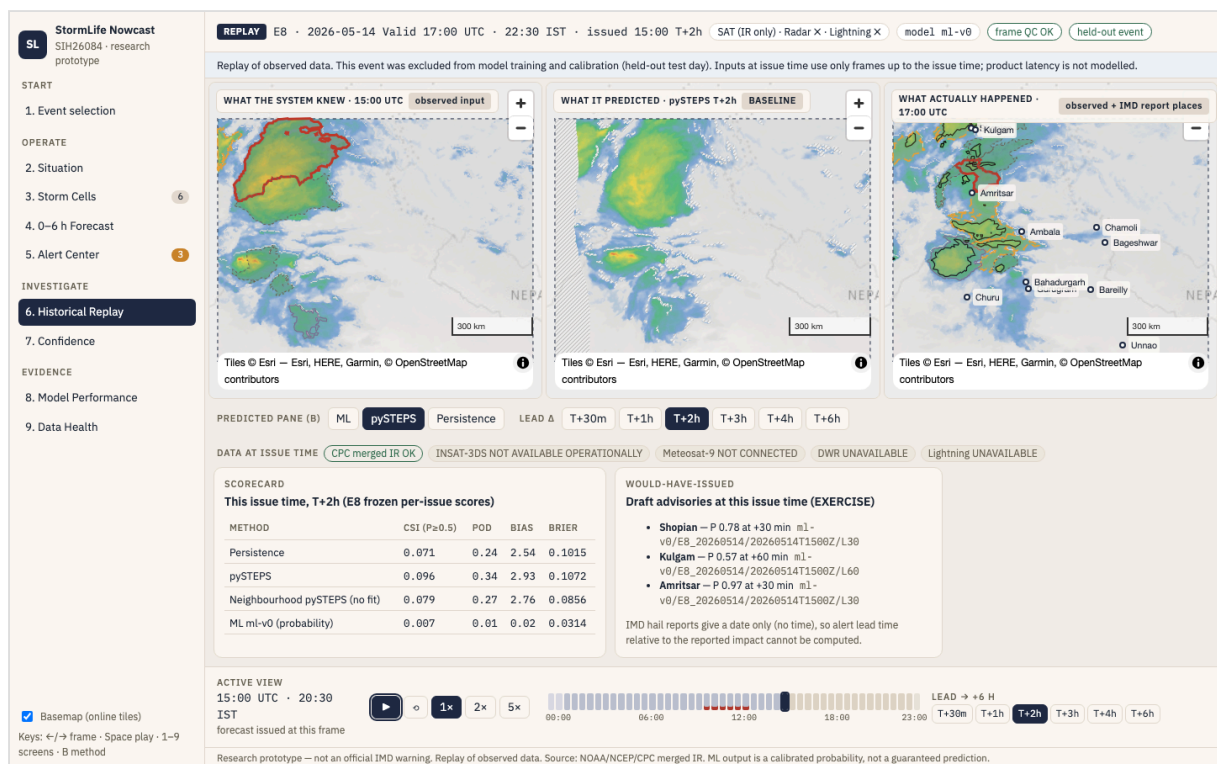


Figure 16. Historical Replay screen, E8: the three synced panes with the pySTEPS forecast at T+2h.

21. Data health and uncertainty

Element	What is shown	Source
QC failures	Per-frame strip (48 frames): OK or QC-fail, bar height = residual missing fraction; E8 has 6 QC-fail frames (10:00–12:30 UTC), kept and flagged	Cube qc_status, qc_missing_frac

Element	What is shown	Source
Source status	CPC merged IR AVAILABLE (FALLBACK); Meteosat-9 and INSAT NOT CONNECTED; IMD DWR and lightning UNAVAILABLE; IMD hail report AVAILABLE (context only)	DATA_REALITY; bundle health.json
Confidence	Skill ribbon at every lead; leads beyond validated skill de-emphasised (baselines: useful FSS 40 km; ML: pooled BSS > 0, with a "small" warning when BSS < 0.1)	skill.json
Calibration / reliability	Reliability diagrams and ECE per lead bin (Table 13)	reliability.csv
Uncertainty	Within-event block-bootstrap intervals per event (Table 12)	bootstrap_ml_vs_pysteps.csv
Scorable area	Unscorable fraction by lead (E8: 9 % at 30 min → 35 % at 6 h)	Phase-2 metrics
Model card	ml-v0: algorithm, 216 trees, train/validation days, held-out events, threshold $p \geq 0.5$, artifact hashes	model_card.json
Limitations	Listed on screen and in FINAL_CLAIMS \$D	health.json
Hashes / provenance	Raw-file SHA-256 per frame; bundle input hashes (verified against the freeze manifest); freeze-manifest hash	manifest.json, docs/FREEZE_ml-v0.json

Table 17. Data-health and uncertainty elements.

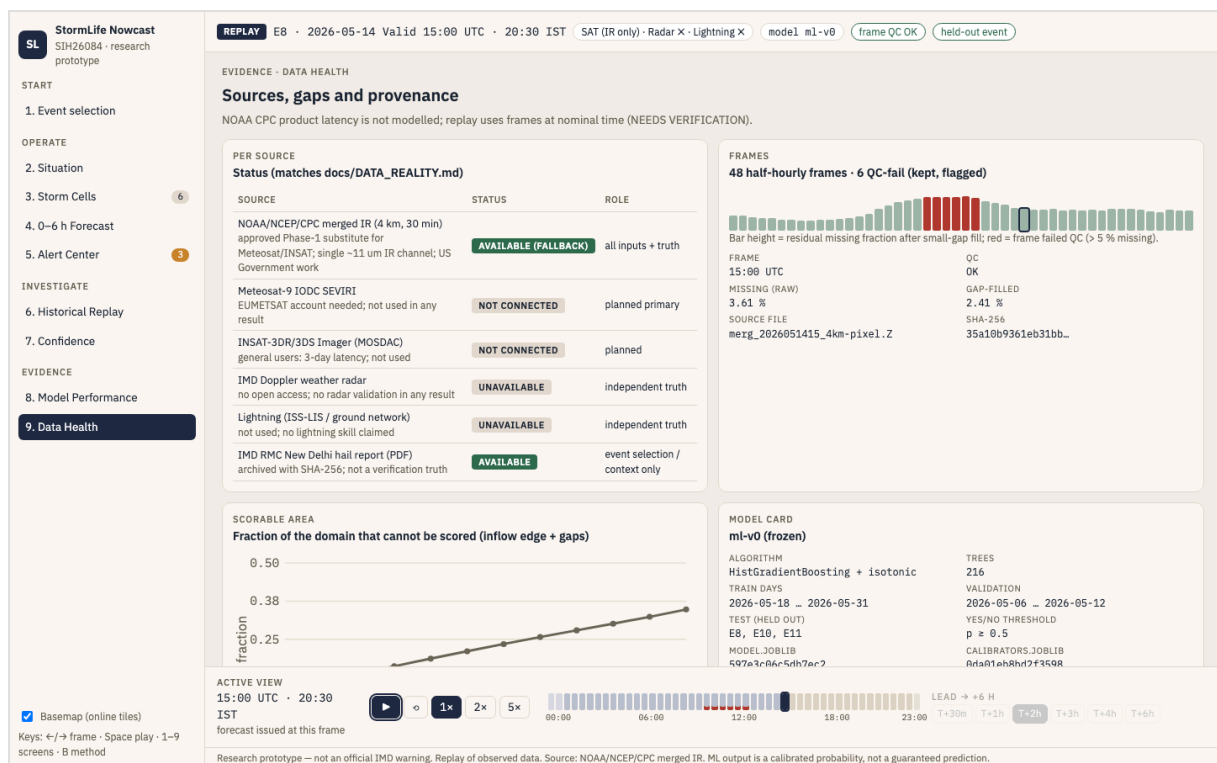


Figure 17. Data Health screen: source status, QC strip, model card.

22. Testing and reproducibility

Suite	Scope	Final result
Python, tests/test_phase1.py	Decoding (scaling, missing, row order), source/target grids, movement, QC flags, phase rules v0, raw-file checksums, cube timing, physical tracks	10 tests
tests/test_phase2.py	Scores, FSS = pySTEPS FSS, leakage (future frames destroyed), baseline hashes	7 tests
tests/test_phase3.py	Overlap linking, merge/split once, threshold, causality, centroid, E8 v2 validity, Phase 1–2 unchanged	13 tests

Suite	Scope	Final result
tests/test_phase4.py	Bootstrap, event selection rule, raw files = manifest, identical protocol, identical pixels, recomputed scores, provenance	16 tests (1 skipped: excluded event E8a)
tests/test_phase5.py	Exact bias factorisation, re-scoring = Phase 4, oracle, lifecycle gate STOP, causality	8 tests
tests/test_phase6.py	Split/leakage rules, feature causality, sampling weights, dataset re-sampling, training reproducibility, calibration on validation, baselines = Phase 4, identical pixels, one held-out forecast recomputed, download resume, window builder	13 tests
tests/test_phase7_freeze.py	All frozen files unchanged; frozen model = evaluated model	2 tests
tests/test_phase8_bundle.py	Bundle inputs = frozen files; output hashes; displayed skill = frozen tables; scores = per-issue table; unique exercise alerts; no future information; tiles reproduce ML probability; event index	8 tests
Python total	<code>.venv/bin/python -m pytest -q</code>	76 passed, 1 skipped
Web unit (Vitest)	Colour scales, tile decoding, CAP exercise format, time formatting	6 / 6 pass
Browser end-to-end (Playwright, headless Chromium, production build)	Full E8 demo flow with no page errors and frozen numbers on screen; E10 opens; unavailable events cannot be opened	2 / 2 pass
Build	<code>npm run build</code> (TypeScript type-check + Vite production build); fresh <code>npm ci</code> from the lockfile	Pass
Local-run verification	Browser pass with the basemap on: map-click selection, playback, keyboard navigation, CAP download, E10/E11, 1280-px layout	0 console errors, 0 page errors, 0 failed local requests, 146/146 map tiles
Freeze manifest	<code>docs/FREEZE_ml-v0.json</code> : 386 files, SHA-256 <code>b5c48807bab48dcc...</code> , frozen 2026-09-25T07:26:44 UTC	Freeze guard passes
Byte-identical reruns	Baselines (Phase 2), v2 tracks (Phase 3), cross-event tables (Phase 4), reliability gate (Phase 5), dataset days and training (Phase 6)	Verified
Secret scan	Not performed in Phases 1–9. Documented controls only: no credentials used or stored (no MOSDAC/EUMETSAT accounts); <code>.env</code> is git-ignored	Manual item

Table 18. Test and reproducibility summary. Sources: SUBMISSION_CHECKLIST §7; PROVENANCE; test files.

Isolation. The prototype needs no server-side state. The bundles are static files, alert approvals stay in the viewer's browser storage, and a fresh `npm ci` + build from the lockfile was verified.

23. Running the prototype

Prerequisites.

- Python 3.11 virtual environment at `.venv` with `requirements.txt`. pySTEPS 1.21.5 must be built from the PyPI source with `-fopenmp` removed on macOS arm64 (BASELINE_EVAL §9).
- Node.js ≥ 18 (verified with 26.7.0) and npm.
- Internet for the basemap and fonts. Both are optional: untick *Basemap* for offline use.

Commands (from the repository root), as verified in the local-run check:

```
# start the demo (bundles and npm packages already present)
cd apps/web && npm run build && npm run preview          # -> http://localhost:4173/

# fresh checkout: rebuild bundles only if apps/web/public/bundles/ is missing (~4 min), then
install
.venv/bin/python -m pipelines.replay.build_bundle --event E8_20260514 E10_20260504
E11_20260516
cd apps/web && npm ci && npm run build && npm run preview

# tests
.venv/bin/python -m pytest -q                          # Python suite incl. freeze guard
cd apps/web && npm test                                  # web unit tests
cd apps/web && npx playwright install chromium && npm run e2e  # browser tests
```

Notes.

- Port 4173 must be free; otherwise Vite uses the next port, so read the printed URL.
- Use a fresh private window for a clean demo (Alert Center approvals persist in browser storage).
- No other deployment (server, container, cloud) exists or is claimed.

24. Use cases / impact

These are **intended uses of a research prototype**. None has been field-tested, and no operational use by any government agency is claimed.

Use case	What the prototype supports today	Evidence level
Meteorological situational awareness	Satellite IR situation map with tracked cold cells and motion, and a probability map of deep convection with its validated skill	Validated on 3 held-out days (grid)
Storm monitoring	Per-cell card, history and advected track; top-5 cells by ML risk	Diagnostic (cell skill only to 30 min)
Disaster-management decision support	Place-level draft advisories for human review; CAP 1.2 export (status=Exercise)	Exercise only; alert skill not verified
Historical event analysis	Replay of real days (<i>knew / predicted / happened</i>) with method switching	Implemented
Research / validation	Reproducible test bed: same baselines, same pixels, frozen evidence, bootstrap intervals; any new model can be scored the same way	Implemented

Table 19. Supported use cases. Coverage potential: geostationary IR covers all of India every 30 min, including areas without radar. The prototype was validated only over NW India.

25. Limitations

Limitation	Current impact	Future improvement
Only 3 held-out event days (E8, E10, E11)	No cross-event significance (best sign test $p = 0.125$); E11 weak (base rate 0.4 %)	Multi-season, multi-region event set
May 2026, NW India only	Nothing shown for monsoon, cloudbursts, other regions or years	Monsoon and other-region events
Backward-in-time ML test	Training (18–31 May) follows the test days; ± 1 -day buffers limit, but do not rule out, weather-regime overlap	Forward test on new days after 31 May, or on pre-2026 archives
Thin training data	14 training days, 7 convective (exactly the gate minimum)	Longer archive
30-minute cadence	No 15-min leads; motion from 60 min of history	15-min Meteosat-9 / INSAT-3DS
Single IR source / channel	No water-vapour, visible or tri-spectral cues for initiation	Multi-channel satellite input

Limitation	Current impact	Future improvement
Source substitution	NOAA CPC merged IR stands in for Meteosat/INSAT; satellite-ID code unverified	Connect the approved sources
Parallax	Not corrected; product status unverified; cloud-top positions may be displaced	Parallax correction / verification
No independent radar or lightning validation	Truth is the same satellite field; tracking and hazard claims unvalidated	IMD DWR, NRSC/IITM lightning where access allows
Edge effects	Long-lead baseline scores cover a shrinking region (E8 unscorable 9 → 35 %); pySTEPS has no inflow forecast	Larger motion domain than scoring domain
Tracking uncertainty	Area jumps ≈ 28 %; 21–35 % direction reversals; phase rules v0 unvalidated	Manual labels / radar tracks; better segmentation
ML binary performance	At $p \geq 0.5$, no gain over pySTEPS beyond 60 min; CSI 0 from 150 min	Validation-chosen operating point (new version, new test days)
ML mechanism	Mostly calibrated smoothing of pySTEPS; initiation skill not shown	Initiation-specific target and score
No live feed	No latency model, no real-time ingestion	Live pipeline with latency model
Prototype / replay only	Static bundles; exercise alerts only; not an official warning	Operational design with IMD/NCMRWF
Hazards	No hail, lightning, rain, downburst or cloudburst skill	Hazard-specific truth and models
Basemap	Esri terms and boundary depiction for a public deployment unverified	Approved basemap (e.g. Survey of India compliant)

Table 20. Limitations. Sources: FINAL_CLAIMS §C–§D; MULTI_EVENT §9; ML_PROTOTYPE §7; TRACKING_AUDIT; FIRST_EVENT.

26. Future scope

All items below are NOT CURRENTLY IMPLEMENTED.

1. **Forward-in-time validation** on storm days after 31 May 2026, or on pre-2026 archives (e.g. GPM_MERGIR via Earthdata; access not verified).
2. **Additional Indian satellite sources.** INSAT-3DR/3DS via MOSDAC and Meteosat-9 IODC via EUMETSAT, with multi-channel input (water vapour, split-window, visible).
3. **Higher temporal resolution:** 15-min imagery and 15-min leads.
4. **Radar and lightning integration** as independent truth and as inputs (IMD DWR; NRSC LDSN / IITM ILLN), where access allows.
5. **Improved storm-initiation modelling**, with an initiation-specific score (VALIDATION_PLAN §2, CI objects).
6. **Lifecycle modelling, only if justified.** Revisit only if more reliable tracking truth makes lifecycle tendencies persistent; the Phase-5 gate currently says STOP.
7. **Operational data pipeline:** live ingestion with a latency model (REPLAY_SPEC §3), API (FastAPI) and storage (PostGIS), as designed in ARCHITECTURE.
8. **Uncertainty-aware forecasting.** Conformal intervals for cell ETAs (VALIDATION_PLAN §6), and a validation-chosen probability operating point evaluated on new held-out days.
9. **Larger multi-season validation:** monsoon events, cloudburst cases, other regions, and bootstrap by event day with enough events for significance.

27. Conclusion

The project delivers a **reproducible, satellite-first nowcasting prototype** for SIH26084, built entirely on real, openly obtainable data.

What was shown, on three held-out hail-report days in May 2026 over NW India:

- **Baselines:** motion extrapolation (pySTEPS) consistently beats persistence, and useful extrapolation skill ends at about 2 h.
- **Error diagnosis:** the growing over-forecast at long leads is mostly a scoring-edge effect.
- **ML:** a lightweight, leakage-safe ML model gives **the best calibrated probability forecasts of deep convection to about 4 h**. It does **not** improve yes/no forecasts beyond 1 h, and it does not demonstrate storm-initiation skill.
- **Tracking:** tracking was audited before use, and cell-level results are reported as diagnostic only.
- **Freeze:** all evidence is frozen, hashed and guarded by tests.
- **Dashboard:** a working replay dashboard shows each forecast with its validated skill, data health and limitations. Advisories are exercise drafts that require human approval.

Limits stated alongside the results: few events, a backward-in-time ML test, a single 30-min IR channel, unverified parallax, and no independent radar or lightning truth.

28. References

References are those already documented in the project (research report, VALIDATION_PLAN, PROVENANCE). No other sources were used.

1. SIH 2026 Problem Statements portal, PS SIH26084. <https://sih.gov.in/sih2026PS>
2. SIH 2026 Guidelines. <https://sih.gov.in/letters/2026/SIH%202026%20Guidelines.pdf>
3. SIH 2026 Idea Presentation Format. <https://sih.gov.in/letters/2026/SIH2026-IDEA-Presentation-Format.pptx>
4. NOAA/NCEP/CPC Globally Merged IR (4 km, 30 min). https://ftp.cpc.ncep.noaa.gov/precip/global_full_res_IR/
5. IMD RMC New Delhi, *Hail storm report of the Northwest India* (updated 16 May 2026). https://mausam.imd.gov.in/newdelhi/mcdata/hailstorm_report.pdf
6. Pulkkinen, S. et al. (2019). pysteps: an open-source Python library for probabilistic precipitation nowcasting. *Geosci. Model Dev.* 12, 4185. <https://gmd.copernicus.org/articles/12/4185/2019/> · <https://github.com/pySTEPS/pysteps>
7. Heikenfeld, M. et al. (2019). tobac. *Geosci. Model Dev.* 12, 4551. <https://gmd.copernicus.org/articles/12/4551/2019/> · Sokolowsky, G. A. et al. (2024). tobac v1.5. *Geosci. Model Dev.* 17, 5309. <https://gmd.copernicus.org/articles/17/5309/2024/> · <https://github.com/tobac-project/tobac>
8. Dixon, M. & Wiener, G. (1993). TITAN: Thunderstorm Identification, Tracking, Analysis, and Nowcasting. *J. Atmos. Oceanic Technol.* [https://doi.org/10.1175/1520-0426\(1993\)010%3C0785:TITIDA%3E2.0.CO;2](https://doi.org/10.1175/1520-0426(1993)010%3C0785:TITIDA%3E2.0.CO;2)
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11. NWCSAF Rapidly Developing Thunderstorms (RDT-CW) description. https://www.nwcsaf.org/rdt_description_2025
12. MeteoSwiss TRT (Thunderstorms Radar Tracking). <https://www.meteoswiss.admin.ch/about-us/research-and-cooperation/projects/en/2002/trt.html>
13. IMD nowcasting with WDSS-II. <https://link.springer.com/article/10.1007/s00703-014-0315-7> · https://nwp.imd.gov.in/WDSSII_website.pdf
14. MOSDAC data access policy. <https://www.mosdac.gov.in/data-access-policy> · INSAT-3DR/3DS: <https://mosdac.gov.in/insat-3dr>, <https://mosdac.gov.in/insat-3ds>
15. EUMETSAT Data Store, Meteosat IODC SEVIRI (EO:EUM:DAT:MSG:HRSEVIRI-IODC). <https://data.eumetsat.int/product/EO:EUM:DAT:MSG:HRSEVIRI-IODC>
16. DownToEarth: IMD locking up its AWS/ARG data portal. <https://www.downtoearth.org.in/climate-change/imd-locking-up-its-awsarg-data-portal-hampers-public-weather-alerts-experts>
17. Project documents: SIH26084_Research_Report.md, docs/PRD.md, docs/ARCHITECTURE.md, docs/DATA_REALITY.md, docs/VALIDATION_PLAN.md, docs/REPLAY_SPEC.md, docs/UI_UX_SPEC.md, docs/FIRST_EVENT.md, docs/BASELINE_EVAL.md, docs/TRACKING_AUDIT.md, docs/MULTI_EVENT.md, docs/DECAY_AWARE.md, docs/ML_PROTOTYPE.md, docs/FINAL_CLAIMS.md, docs/FINAL_RESULTS.md, docs/PRODUCT_DEMO.md, data/PROVENANCE.md.

29. Appendices

Appendix A — Frozen metrics

Metric	Value	Source file
BSS, ML (30 / 60 / 120 / 180 / 240 / 360 min)	0.67 / 0.51 / 0.27 / 0.13 / 0.06 / −0.03	data/processed/ml_eval/ml-v0/metrics_pooled_lead.csv
BSS, NP31	0.63 / 0.41 / 0.02 / −0.23 / −0.37 / −0.59	same
BSS, pySTEPS	0.44 / 0.10 / −0.38 / −0.67 / −0.82 / −1.11	same
BSS, persistence	0.32 / −0.04 / −0.47 / −0.69 / −0.78 / −0.84	same
CSI ($p \geq 0.5$), ML	0.583 / 0.452 / 0.243 / 0.000 / 0.000 / 0.000	same
CSI, pySTEPS / persistence (pooled)	0.586 / 0.429 / 0.258 / 0.172 / 0.122 / 0.049 · 0.511 / 0.342 / 0.176 / 0.092 / 0.052 / 0.022	same; cross_event_by_lead.csv
Useful FSS 40 km	90 min (P), 120 min (A), all 3 events	decay_by_event.csv
ECE, ML	0.0022 / 0.0034 / 0.0042 / 0.0026	reliability.csv
Bias, E8 6 h	$2.61 = 3.76 \times 0.73 \times 0.96$; whole domain 0.70	phase5/decomposition_by_lead.csv
Tracking, E8	ID switches 0.206 → 0.004; zero-overlap 0.057 → 0; reversals 0.47 → 0.35	audit_v1.json, audit_v2_none.json
Object diagnostic, 30 min	0.59 / 0.62 / 0.57 (E8 / E10 / E11)	object_diagnostics_by_event.csv

Appendix B — Event table

See Table 15.

Event-day exclusions for ML:

- 30 Apr–2 May (E8a);
- 3–5 May (E10);
- 13–15 May (E8);
- 15–17 May (E11).

Frames per day: 48 (00:00–23:30 UTC). **Issue times per day:** 45.

Appendix C — Technology stack

See Table 1. **Not used:** SatPy, rasterio, MetPy, Tailwind CSS, deck.gl, LightGBM, PyTorch, SHAP, xskillscore, FastAPI, PostGIS, Docker.

Appendix D — Test summary

See Table 18.

- **Python:** 77 tests collected (10 + 7 + 13 + 16 + 8 + 13 + 2 + 8), 76 passed and 1 skipped.
- **Web:** 6 unit tests and 2 end-to-end tests pass.
- **Build:** passes.

Appendix E — Reproducibility information

- Raw data: data/raw/cpc_merged_ir/manifest.json (URL, Last-Modified, bytes, SHA-256 per file).

- Configs: `config/domain.yaml`, `tracking.yaml`, `tracking_v2.yaml`, `baseline.yaml`, `phases.yaml`, `events.yaml`, `ml_dataset.yaml`.
- Per-phase provenance: `baseline_provenance.json`, `run_summary.json`, `multi_event_provenance.json`, `decomposition_provenance.json`, `lifecycle_decision.json`, `dataset_manifest.json`, `model_card.json`, `evaluation_provenance.json`, the bundle `manifest.json` files, and `data/PROVENANCE.md`.
- Environment: Python 3.11.16; package versions in Table 1 and in `docs/FREEZE_ml-v0.json`.

Appendix F — Model freeze / provenance

Item	SHA-256
<code>data/models/ml-v0/model.joblib</code>	597e3c06c5db7ec2f6c9e7e847b49a7743949a1a69a01ca20285d56c52f51857
<code>data/models/ml-v0/calibrators.joblib</code>	0da01eb8bd2f35988141c5fdcab03658bcc6e4136bfb2299a85a72a7aacc1bc7
<code>data/models/ml-v0/model_card.json</code>	2a763b0aaeae985834bf22fc47b2e6d0fa367975892759439c94049c2f29728e
<code>config/ml_dataset.yaml</code>	bda7c3859a93474d9329286e905a58f8a76dba6af2acf6bdc848d7e0628fe4f4
Dataset manifest	9e66e49a1485e3bda6c9e051ba58937d43a11bbc1591f67367fab8e5db9459fe
Evaluation provenance	d2dadac4140240f0a9eb255a2d13c86543f88685fa30cbc9efdddc5b50bc32784
<code>docs/FREEZE_ml-v0.json</code> (386 files, frozen 2026-09-25T07:26:44 UTC)	b5c48807bab48dccab1706d6d2bc144260cd4f3249dc533b09647323d653fe35

Source: `data/PROVENANCE.md` §Phase 7.

Appendix G — Demo flow

The verified E8 browser flow (DEMO_VIDEO_SCRIPT, 4:40):

1. Open event **E8 · 14 May 2026**.
2. **Situation at 14:00 UTC**: 9 cells.
3. **Select cell #108**: Developing, 216.1 K.
4. **0–6 h Forecast**: pySTEPS, then **ML probability** at T+2h (ribbon: CSI 0.243 · BSS 0.27 · FSS40 0.53).
5. **Historical Replay**: B toggle and playback, 14:00 → 15:00.
6. **Model Performance** (key 8).
7. **Data Health** (key 9): 6 QC-fail frames.
8. Optional **Alert Center** at 08:00 UTC: approve Shopian, then CAP status=Exercise.

End of report.